



Review article

A review of physics-based low-temperature proton-exchange membrane fuel cell models for system-level water and thermal management studies

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HIGHLIGHTS

- Review of low computational cost physics-based PEMFC numerical models.
- Focus on transport phenomena impacting water and thermal management performance.
- Analysis of 54 models with different levels of detail and modelling hypotheses.
- Comparison of model validation practices and computational time.

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ABSTRACT

Water management and thermal management are essential for the design and operation of low-temperature proton-exchange membrane fuel cells and demand a deep understanding of the fuel cell system behaviour. System-level studies require fast and simple numerical models, a challenge given the complexity of the transport phenomena involved. Numerous models exist in the literature, with different levels of complexity and different hypotheses aiming at simplifying the representation of transport phenomena. This review summarizes different hypotheses and modelling approaches adopted in low computational cost physics-based proton-exchange membrane fuel cell models, with a focus on aspects related to water and thermal management. It also describes modelling approaches for auxiliary system components, validation practices and compare the computational efficiency of different models. We put in evidence the high diversity of hypotheses and modelling approaches adopted in the literature, which reflect the trade-off between computational cost, model complexity and accuracy. We conclude that rather than developing new models from scratch, a strong effort on the validation of the existing models is needed, with the development of a standard validation procedure.

1. Introduction

Fuel cells are devices that allow the conversion of hydrogen and oxygen into electricity, heat and water by electrochemical reactions. Being so, they are essential elements for the use of hydrogen as an energy vector. In particular, low-temperature proton-exchange membrane fuel cells are viewed today as a promising technology for applications in transportation (buses, trucks, trains, auxiliary power units for airplanes) and small-scale distributed power generation, both stationary and portable [1]. However, high cost and low durability of the equipment, as well as integration issues in some vehicle applications still hinder a wider adoption of fuel cells. In order to reduce these barriers, improvements in the design and operation of fuel cell systems are needed.

Water and thermal management of fuel cells are topics where there is room for such improvements. The goal of water management is to keep the proton-exchange membrane hydrated while avoiding excessive water inside the cells [2]. Indeed, the perfluorinated sulfonic acid

membranes used in low-temperature PEM fuel cells need to be hydrated to effectively conduct protons and fulfil their role as an electrolyte. If they dry, their proton conductivity decreases, leading to an increased heat release and a decrease of the fuel cell efficiency. Heat release may lead to a local temperature increase, which will, in turn, intensify the membrane drying in a process leading to the irreversible degradation of the membrane. On the other hand, excessive water inside the cell may lead to condensation and the obstruction of gas transport by liquid water accumulated in the porous layers, a phenomenon known as flooding [3]. Concerning thermal management, its goal is to evacuate the heat generated by the fuel cell while keeping an appropriate and homogeneous temperature of operation. Thermal management of low-temperature fuel cells is particularly challenging as their operating temperature is around 80 °C. It is more difficult to dissipate heat at this temperature than in the case of an internal combustion engine operating around 400 °C [4]. Water and thermal management are intimately connected: an increase of the operating temperature will favour

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Nomenclature**Abbreviations**

AC	Along-the-channel
BP	Bipolar plate
CL	Catalyst layer
CSTR	Continuously-stirred tank reactor
DM	Diffusion medium
EIS	Electrochemical impedance spectroscopy
GC	Gas channel
GDL	Gas diffusion layer
IP	In-plane
MEA	Membrane-electrode assembly
MPL	Micro-porous layer
NTU	Number of transfer units
OCV	Open-circuit voltage
PEM	Proton-exchange membrane
PFSA	Perfluorinated sulfonic acid
RH	Relative humidity
RT	Real time
TP	Through-plane

Greek symbols

α	Transfer coefficient, -
η	Efficiency, -
η	Overpotential, V
γ	Heat capacity ratio, -
λ	Membrane/ionomer water content, -
λ	Stoichiometry, -
μ	Dynamic viscosity, Pa s
ω	Rotational speed, rad s ⁻¹
ρ	Density, kg m ⁻³
σ_l	Liquid water surface tension, N m ⁻¹
σ_{memb}	Membrane conductivity, $\Omega^{-1} \text{ m}^{-1}$
τ	Torque, N m
θ	Contact angle, -
ϵ	Porosity, -
ϵ_p	Percolation threshold, -
ξ	Electroosmotic drag coefficient, -

Roman symbols

$\dot{m}$	Mass flow rate, kg s ⁻¹
$\dot{Q}$	Gas flow rate, m ³ s ⁻¹
$\dot{Q}$	Heat source, W
A	Surface area, m ²
a	Water activity, -
C	Heat capacity, J kg ⁻¹
c	Concentration, mol m ⁻³
c_f	Sulfonic acid concentration, mol m ⁻³
c_p	Specific heat at constant pressure, J kg ⁻¹ K ⁻¹ or J mol ⁻¹ K ⁻¹
D	Diffusion coefficient, m ² s ⁻¹
D_λ	Membrane water diffusion coefficient, m ² s ⁻¹
D_T	Thermoosmotic diffusion coefficient, mol m ⁻¹ K ⁻¹ s ⁻¹
E	Potential, V
F	Faraday's constant, 96 485 C mol ⁻¹
h	Heat transfer coefficient, W K ⁻¹ m ⁻²
h	Specific enthalpy, J kg ⁻¹

i	Current density, A m ⁻²
i_0	Exchange current density, A m ⁻²
K	Permeability, m ²
k_{abs}	Absorption/desorption rate coefficient, m s ⁻¹ or mol m ⁻² s ⁻¹
M	Molar mass, kg mol ⁻¹
N_{cells}	Number of cells in the stack, -
P	Power, W
p	Pressure, Pa
p_c	Capillary pressure, Pa
q	Mass flux, kg m ⁻²
R	Universal gas constant, 8.3145 J mol ⁻¹ K ⁻¹
r	Resistivity, $\Omega \text{ m}$
r_{H^+}	Protonic resistivity, $\Omega \text{ m}$
r_{el}	Electrotonic resistivity, $\Omega \text{ m}$
S	Entropy, J mol ⁻¹ K ⁻¹
s	Liquid water saturation, -
S_k	Mass source of species k , kg s ⁻¹
S_T	Volume heat source, J s ⁻¹ m ⁻³
s_{ice}	Ice saturation, -
T	Temperature, K
t	Time, s
t_i	Thickness of layer i , m
V	Voltage, V
v	Velocity, m s ⁻¹
x	Mole fraction, -

Subscripts and superscripts

0	Standard state
<i>memb</i>	Membrane
<i>act</i>	Activation
<i>amb</i>	Ambiant
<i>an</i>	Anode
<i>ca</i>	Cathode
<i>cm</i>	Compressor motor
<i>conc</i>	Concentration
<i>cool</i>	Cooling
<i>cp</i>	Compressor
<i>eff</i>	Effective
<i>el</i>	Electric
<i>eq</i>	Equilibrium
<i>g</i>	Gas
<i>in</i>	Inlet
<i>i</i>	Layer i
<i>k</i>	k th segment along the channel
<i>k</i>	Species k
<i>l</i>	Liquid
<i>ohm</i>	Ohmic
<i>opt</i>	Optimum
<i>out</i>	Outlet
<i>req</i>	Required
<i>rev</i>	Reversible
<i>st</i>	Stack
<i>th</i>	Thermal
<i>w</i>	Water

water evaporation and membrane drying, while a decrease will favour condensation and flooding. In turn, water evaporation or condensation will absorb or release a significant amount of heat. Water and thermal management are also related to the startup and operation of fuel cells at freezing temperatures, where the solidification of liquid water inside the cell may bring several issues [5].

Water and thermal management of fuel cells require or are impacted by a number of peripherals that contribute significantly to the fuel cell system cost [6]. Those are part of the air supply, fuel supply and cooling sub-systems. We can cite as examples:

- air supply/cathode sub-system: compressor or blower, humidifier, back-pressure valve;
- fuel supply/anode sub-system: hydrogen recirculation pump (or ejector), water separator, purge valve;
- cooling sub-system: cooling pump, three-way valve, radiators.

Air humidifiers and radiators, in particular, can be voluminous and bring integration difficulties. Valves, air compressor/blower and pumps must be controlled in a way that optimizes performance and durability of the fuel cell under the different expected operating conditions [7]. System-level studies, where all those peripherals, their behaviour and their interactions are taken into account are therefore fundamental for improving the design and operation of fuel cell systems from a water and thermal management standpoint.

But such system-level studies require numerical models able to simulate the behaviour of the fuel cell system with a sufficiently low computational cost. Indeed, one should be able to simulate with a reasonable computational time the operation of the fuel cell system over time intervals corresponding to load cycles encountered in potential applications, such as vehicle driving cycles. These may last from several minutes to several hours. Moreover parametric studies and optimization algorithms require running simulation cases several times, with the computational cost increasing even more.

Detailed models for PEM fuel cells are, however, too computationally expensive for those kind of studies. Indeed, these detailed models are often continuum models based on the solution of conservation equations for mass, momentum, species (H_2 , O_2 , N_2 and H_2O in its different phases), heat and charge (both positive and negative) [1,8]. These equations take the form of partial differential equations, in some cases containing highly non-linear terms, which must be discretized and solved in both time and space. Even solving such a detailed model on only two dimensions such as in the case of Goshtasbi et al. [9] can take, depending on the conditions, between 5 and 12 h to simulate 340 s of the PEM fuel cell behaviour on a desktop computer (that means between 50 and 130 times slower than real time). The 2D+1D model of Nandjou et al. [10] took 6 h to calculate a single steady-state operating point.

But the computational cost is not the only drawback of such detailed models. Complexity of the model also brings difficulties into its use in fuel cell system development. The more complex the model is, the more time is needed to implement and maintain it, to setup simulation cases, to debug errors and to post-treat and interpret results. Further, complex models often require a long list of model parameters, some of which have been obtained for conditions or materials different from those under study, some of which are highly uncertain and some of which are not available at all, thus requiring potentially long and costly experimental work. Besides, uncertainty in model parameters can easily curtail the accuracy gained by using a more complex model. The works by Jiang et al. [11] and Vetter and Schumacher [12,13] are particularly enlightening on how uncertainty on key parameters for PEM fuel cell models can have a strong impact on the results. For all those reasons, complex models are not suited to develop fuel cell systems in an agile way.

Obtaining faster and simpler models for PEM fuel cells can therefore accelerate the PEM fuel cell system development. But it requires simplifying hypotheses, since the phenomena involved in the fuel cell

operation are complex and encompass multiple scales in space and time. The literature on PEM fuel cell modelling contain plenty of proposals of simplified models that can be used for system-level studies. Those models adopt a wide variety of hypotheses and approaches that aim to limit the number of equations, the number of points on which the solution is sought (both in space and time dimensions), the evaluation of computationally-intensive non-linear functions or the coupling between different phenomena involved. Many of these hypotheses concern the transport of reactant gases, water in its different phases and heat inside the fuel cells. As a consequence, the impact of these hypotheses is of great importance for those interested in water and thermal management of fuel cells. In spite of that, to the author's knowledge, little has been done to identify the impact of each individual hypothesis on the accuracy of the model and the reduction of the computational cost, as well as the domain of validity of each hypothesis. Due to the complexity of the phenomena involved and the variety of hypotheses and models encountered in the literature, undertaking that analysis in a systematic way is not an easy task. In this review, we propose a first step towards that goal: a presentation of the different hypotheses adopted in system-level and/or control-oriented PEM fuel cell models available in the literature.

In the last ten years, several review papers concerning fuel cell modelling have been published. The review of Weber et al. [14] focused on the modelling of transport phenomena in PEM fuel cells, with a very good presentation of the basic governing equations for mass, charge and energy transport and of phenomena occurring in the catalyst layer, for which the reader is referred. The review of Andersson et al. [8] focused on the modelling of multi-phase flow in the gas diffusion layers and gas channels, presenting different computational approaches at the cell scale and a valuable discussion on the different transport properties in the GDL. The reviews of Jiao's group on water transport [2], cold start [5], gas diffusion layers [15] and thermal management [4] also included some modelling aspects and are an important starting point for those interested in water and thermal management of PEM fuel cells. The review of Jahnke et al. [7] was more general and covered the different modelling scales from atomistic to system level, with a focus on the modelling of degradation phenomena. More recently, Zhao et al. [16] reviewed fuel cell modelling for real-time control, looking into both low-dimension (from 0D to 1D+1D) physics-based models and data-driven models. As Zhao et al., we focus on low-dimension models which can simulate the fuel cell performance with lower computational cost, allowing for system-level and/or control-oriented studies. But more specifically, we concentrate our review on modelling aspects and hypotheses related to the study of fuel cell water and thermal management.

2. Modelling the PEM fuel cell

Fuel cell models are often continuum (macroscopic) models based on conservation equations [1,8]. Continuum models are not necessarily suited to the modelling of the fuel cell, as the length scale of the representative elementary volume for the porous gas diffusion layer is close to the thickness of this layer [17,18]. Pore-scale (microscopic) modelling is an alternative allowing for a detailed description of liquid water transport at the microscopic scale and the obtainment of averaged macroscopic transport properties. Pore-scale modelling include the lattice Boltzmann method (LBM) [19–22], the volume-of-fluid (VOF) method [23–25] or pore network models (PNM) [17,26]. With the current computational capabilities, the computational cost of pore-scale models is prohibitive for the modelling of a full fuel cell and, *a fortiori*, of a full fuel cell system, which can only be handled by continuum models [2]. Being so, a detailed description of pore-scale models is out of the scope of this work. For more details on pore-scale modelling methods, the reader is referred to the review of Chen et al. [18].

Fuel cell continuum models can be classified according to the spatial dimensions considered. The so-called 0D models consider the anode and the cathode as a single control volume each behaving like a continuously-stirred tank reactor, without any spatial consideration. A typical 0D model is schematized in Fig. 1 extracted from Ritzberger et al. [27]. 0D models can use empirical or semi-empirical correlations to overcome the lack of spatial detail, but they are not able to describe spatial phenomena occurring inside the anode or the cathode. Further, 0D models have limited prediction capability as they must often be calibrated for a specific material, under specific operating conditions.

1D models adopt a spatial discretization with more than one control volume along either the flow channel (along-the-channel) direction or the through-plane direction of the cell. These discretizations allow the consideration of transport phenomena, but increase the computational cost. Through-plane 1D models focus on the transport phenomena occurring across the different layers of the MEA and allow to establish through-plane profiles of quantities such as gas concentrations, temperature or liquid water saturation. They contribute to a better understanding of these phenomena, as well as the impact of operating conditions and material properties. Along-the-channel 1D models focus on the effects of the changes in flow conditions in the gas flow channels as reactants are consumed and product water evacuated, and specially on the impact of these changes on the local current density. They often assume that the flow channel can be modelled as a straight channel. Along-the-channel 1D models can be instructive on the impact of flow direction (cross-flow or co-flow).

1D+1D models combine the advantages of both along-the-channel and through-plane 1D models. They have more than one control volume for each of the flow channel and through-plane directions, but without any flux in the in-plane direction of the porous layers, taking advantage of the planar geometry of the cell and the much smaller dimension in the through-plane direction. By doing so, 1D+1D models decouple the solution of adjacent through-plane transport equations, leading to faster simulations than with equivalent 2D models where transport in the in-plane direction is solved. The scheme of the 1D+1D model proposed by Yang et al. [28] is presented in Fig. 2. 2D+1D models work in the same way but consider two-dimensional gas flow channel arrangements. That allows to investigate the impact of different flow fields on the performance of the fuel cell. 2D models take into account in-plane fluxes, which can be important for serpentine and interdigitated flow fields due to a significant pressure gradient between adjacent channels [2]. They often combine the through-plane direction with either the along-the-channel or the across-the-channel direction. In the latter case, the model focus on a cross-section of the flow and the effects of the ribs on the in-plane transport. Finally, 3D models allow for a full representation of the transport phenomena at the cell scale, including through-plane and in-plane transport, as well as the effect of the flow field configuration, but at the expense of a higher computational cost.

Fuel cell models are also classified as transient or steady-state depending on the consideration of dynamic effects. Steady-state models neglect transient effects, which allows for great simplifications in the modelling process and a lower computational cost. Such models can be useful to investigate or optimize the fuel cell performance at key operating points. On the other hand, steady-state models cannot reproduce transient operation phases, such as the fuel cell startup, or the effect of variable operating conditions. Further, these models cannot be used to study control strategies, which requires transient models. The so-called quasi-steady approach use steady-state models which are solved for each time step. The underlying hypothesis is that the dynamic effects considered have a characteristic time much smaller than the chosen time step. Therefore, these dynamic effects are fast enough for the fuel cell to reach steady-state at each time step. Concerning the slower dynamic effects, the steady-state model can be coupled with a transient model which takes them into account. As an example, this was done by Xu et al. [29] where a steady-state model for the fuel cell was integrated

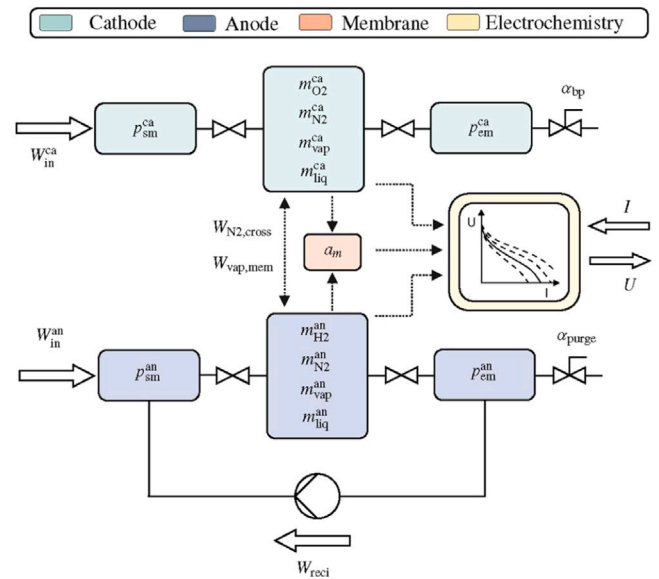


Fig. 1. Schematics of the 0D model used by Ritzberger et al. [27].
Source: Reproduced from Ritzberger et al. [27].

in a system model describing the transient behaviour of the auxiliary components.

In 2004, Pukrushpan et al. [30] proposed perhaps the first control-oriented 0D fuel cell system model taking into account the transient effects of oxygen and hydrogen partial pressures, stack temperature and membrane water content on the stack voltage. The authors clearly stated that their model "is useful in showing the behaviour of fuel cell systems qualitatively, rather than quantitatively". Several works presented control-oriented models in the following years [31–36] while paying increasing attention to the control of the stack temperature and the membrane water content. In particular, del Real et al. [31] developed a two-phase 0D model which was able to reproduce the transient behaviour of a Ballard Nexa 1.2 kW PEM fuel cell system. In the same period, several steady-state models were also developed with along-the-channel [37], through-plane [38,39] or even 2D discretizations, as they allowed to account for spatial dimensions and heat transfer effects which were neglected in 0D models. With time and the improvement of computational capabilities, some models started to combine both detailed spatial discretization and dynamic operation [28,40–44]. Models also started to include two-phase effects more often [11,28,29,36,40,43,45–50], although those were already present in the model proposed by del Real et al. [31]. More recently, some authors came back to simpler models (0D and/or steady-state) in order to reduce the computational time and make faster system-level optimization [49,51], sensitivity analyses [11,52] and model calibration [27,53]. In some cases, analytical solutions of governing equations are included to further reduce computational time [11,29].

Table 1 lists the PEM fuel cell models mentioned in this work and how they could be classified according to the consideration of dynamic effects (steady-state or transient models), the spatial dimensions (from 0D to 3D), as well as the consideration of two-phase phenomena and of temperature heterogeneity (isothermal or non-isothermal). In the following, we will see how those models represent the cell voltage, the transport phenomena in the different layers of the membrane-electrode assembly and in the gas flow channels and associated spatial effects at the cell and stack scales.

2.1. Cell voltage

Most often, the goal of fuel cell models is to predict the operating voltage for a given current density. Models calculate the cell voltage

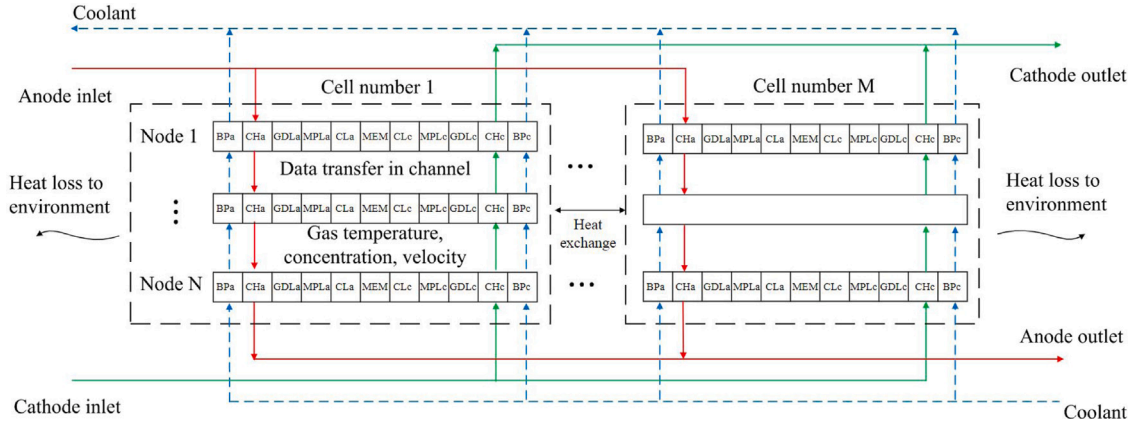


Fig. 2. Schematics of the 1D+1D model from Yang et al. [28].

Source: Reproduced from Yang et al. [28] with permission from Elsevier.

from the current densities either by solving charge conservation equations or by adopting semi-empirical parametrizations. For those solving charge conservation equations, both proton and electron conservation are considered in the through-plane direction. Proton conservation is solved in the catalyst layers and membrane and electron conservation in the catalyst layers, DM and bipolar plates. Examples of models considering charge conservation equations are found in [42,50,56,57,74,76]. In the majority of the models reviewed in this work, the semi-empirical parametrization of the cell voltage is used, for its simplicity and computational efficiency. Further, its parameters can be more easily fitted to experimental data. The cell voltage can be obtained by subtracting activation, ohmic and concentration overpotentials terms from the reversible potential :

$$V_{cell} = E_{rev} - \eta_{act} - \eta_{ohm} - \eta_{conc} \quad (1)$$

The reversible potential is calculated with the Nernst equation:

$$E_{rev} = E_{rev}^0 + \frac{\Delta S^0}{2F}(T - T_0) + \frac{RT}{2F} \ln \left[\left(\frac{p_{O_2}}{p_0} \right)^{0.5} \left(\frac{p_{H_2}}{p_0} \right) \right] \quad (2)$$

with the standard conditions $p_0 = 1$ atm, $T_0 = 298.15$ K, $E_{rev}^0 = 1.229$ V and $\Delta S^0 = 165$ J mol⁻¹ K⁻¹.

2.1.1. Activation overpotential

The activation overpotential is generally represented by a Butler-Volmer formulation [41,42,50,57,76]:

$$i = i_0 \left\{ \exp \left[-\frac{2\alpha F \eta_{act}}{RT} \right] - \exp \left[-\frac{2(1-\alpha) F \eta_{act}}{RT} \right] \right\} \quad (3)$$

or the simpler Tafel equation [28,29,36,38–40,43,45–49,51,52,55,56,63,65,66,72,74,77], valid for large values of η_{act} and negligible mass transfer resistance, given by:

$$\eta_{act} = \frac{RT}{2\alpha F} \ln \left(\frac{i}{i_0} \right) \quad (4)$$

The exchange current density i_0 can be a constant or a function of the temperature and the oxygen concentration. A crossover current (also called internal current) term i_n can be added to the cell current density in Eq. (4) to account for the crossover of gas species across the membrane, yielding the form:

$$\eta_{act} = \frac{RT}{2\alpha F} \ln \left(\frac{i + i_n}{i_0} \right) \quad (5)$$

This crossover current term can be either empirical as in [51,52,63] or calculated from the fuel crossover flux as in [43,66]. As an alternative to the Tafel equation, Jiang et al. [11] solved the charge conservation equations analytically to obtain a parametrization for the activation overpotentials. The semi-empirical parametrization of the activation

overpotential proposed by Amphlett et al. [83] is also extensively used in fuel cell models [33,37,54,59–62,69,73,78–81]. It describes the activation overpotential as a linear combination of four terms which can be fitted to experimental data:

$$\eta_{act} = \xi_1 + \xi_2 T + \xi_3 T \ln(c_{O_2}^*) + \xi_4 T \ln i \quad (6)$$

where the oxygen concentration at the cathode gas-liquid interface $c_{O_2}^*$ is obtained from the Henry's law and ξ_i are the fitting coefficients. Nevertheless, other alternative parametrizations were also used by different authors [30–32,34,44,53,67,75]. Very often the anode activation overpotential is neglected as the hydrogen oxidation reaction is much faster than the oxygen reduction reaction.

2.1.2. Ohmic overpotential

The ohmic overpotential is due to both protonic and electronic resistances across the different layers of the cell and can be written:

$$\eta_{ohm} = r_{ohm} i = (r_{H^+} + r_{el}) i \quad (7)$$

Protonic resistances occur in the catalyst layers and the membrane, while electronic resistances occur across the catalyst layers, the MPL and GDL and the bipolar plates. The protonic resistance is generally dominant and is a strong function of the membrane water content in the perfluorinated sulfonic acid membranes used in low-temperature PEMFC. Most models [11,29,30,34,38–42,45,47–50,55–57,60,62,63,65,66,73,74] still use the empirical relation for the conductivity of a Nafion 117 membrane obtained back in 1991 by Springer et al. [84]:

$$\sigma_{memb} = (0.005139\lambda - 0.00326) \times \exp \left[1268 \left(\frac{1}{303} - \frac{1}{T} \right) \right] \quad (8)$$

with σ_{memb} in $\Omega^{-1} \text{cm}^{-1}$ and T in Kelvins. Some authors multiplied Eq. (8) by a tuning factor [34,48] or fitted its different parameters to their own experimental data [44,53,67]. Mann et al. [85] proposed some correction terms to Eq. (8) that are adopted by several works [54,59,64,73,78–81]. Other less commonly used parametrizations are the one proposed by Amphlett et al. [83] and used by [33,37], defined as:

$$\sigma_{memb} = \xi_5 + \xi_6 T + \xi_7 i \quad (9)$$

or a more complex equation proposed by Weber and Neumann [86] and used by [75,76]. Some specific parametrizations can also be found in [43,51,52]. Given the membrane conductivity, its protonic resistance can be calculated based on the membrane thickness. Calculating the protonic conductivity of the catalyst layer require the knowledge of catalyst properties such as the ionomer volume fraction, catalyst layer thickness and tortuosity. This was done by a few works [29,40–43,49,50,56,57] but most neglected the catalyst layer protonic conductivity.

Table 1
Classification of the main models reviewed in this work.

Pukrushpan et al., 2004	[30]	Transient	0D	Single-phase	Isothermal
Khan and Iqbal, 2005	[54]	Steady-state	0D	Single-phase	Isothermal
Bao et al., 2006	[55]	Transient	1D	Two-phase	Isothermal
Del Real, 2007	[31]	Transient	0D	Two-phase	Isothermal
Huisseune et al., 2008	[37]	Steady-state	1D (AC)	Single-phase	Non-isothermal
Vasu et al., 2008	[32]	Transient	0D	Single-phase	Isothermal
Falcão et al., 2009	[38]	Steady-state	1D (TP)	Single-phase	Non-isothermal
Gerteisen et al., 2009	[56]	Steady-state	1D (TP)	Two-phase	Non-isothermal
Jiao and Li, 2009	[57]	Transient	3D	Two-Phase	Non-isothermal
Karnik et al., 2009	[58]	Transient	0D	Single-phase	Isothermal
Kim et al., 2010	[59]	Transient	0D	Single-phase	Isothermal
Mangold et al., 2011	[35]	Transient	0D	Two-phase	Isothermal
Schultze and Horn, 2011	[60]	Transient	0D	Two-phase	Isothermal
Ziougou et al., 2011	[33]	Transient	0D	Two-phase	Isothermal
Damour et al., 2012	[61]	Transient	0D	Single-phase	Isothermal
Hosseinzadeh et al., 2013	[62]	Steady-state	0D	Single-phase	Isothermal
Chaudhary et al., 2014	[50]	Transient	2D	Two-phase	Non-isothermal
Kim et al., 2014	[39]	Steady-state	1D (TP)	Two-phase	Isothermal
Liso et al., 2014	[34]	Transient	0D	Two-phase	Isothermal
Strahl et al., 2014	[36]	Transient	0D	Two-phase	Isothermal
Zhou et al., 2014	[40]	Transient	1D (TP)	Two-phase	Non-isothermal
Fly, 2015a	[63]	Transient	0D	Two-phase	Isothermal
Kang, 2015	[47]	Transient	2D+1D (TP)	Two-phase	Non-isothermal
Kang and Min, 2016	[48]	Transient	1D (TP)	Two-phase	Non-isothermal
Robin et al., 2015	[44]	Transient	2D+1D (TP)	Two-phase	Non-isothermal
Zhao et al., 2015	[64]	Transient	0D	Single-phase	Isothermal
Abdin et al., 2016	[65]	Transient	0D	Single-phase	Isothermal
Goshdashti et al., 2016	[66]	Transient	1D+1D	Two-phase	Non-isothermal
Headley et al., 2016	[67]	Transient	1D (AC)	Single-phase	Isothermal
Liso et al., 2016	[68]	Steady-state	1D (TP)	Two-phase	Isothermal
Salva et al., 2016	[45]	Steady-state	1D (TP)	Two-phase	Non-isothermal
Schultze and Horn, 2016	[69]	Transient	0D	Single-phase	Isothermal
Tang et al., 2017	[41]	Transient	1D+1D	Two-phase	Non-isothermal
Futter et al., 2018	[42]	Transient	2D	Two-phase	Non-isothermal
Jiang et al., 2018	[11]	Steady-state	1D (TP)	Two-phase	Non-isothermal
Qin et al., 2018	[70]	Steady-state	2D (IP)	Single-phase	Non-isothermal
Sun et al., 2018	[71]	Transient	0D	Single-phase	Isothermal
Wang et al., 2018	[72]	Transient	1D+1D	Two-phase	Non-isothermal
De Campo, 2019	[73]	Transient	0D	Two-phase	Isothermal
Han et al., 2019	[74]	Steady-state	1D (TP)	Single-phase	Isothermal
Lazar et al., 2019	[75]	Transient	1D (TP)	Single-phase	Isothermal
Vetter and Schumacher, 2019	[76]	Steady-state	1D (TP)	Two-phase	Non-isothermal
Yang et al., 2019	[28,77]	Transient	1D+1D	Two-phase	Non-isothermal
Chen et al., 2020	[78]	Transient	0D (scalable)	Two-phase	Non-isothermal
Goshdashti et al., 2020	[43]	Transient	1D+1D	Two-phase	Non-isothermal
Hoeflienger and Hofmann, 2020	[51]	Steady-state	0D	Single-phase	Isothermal
Ritzberger et al., 2020	[27,53]	Transient	0D	Two-phase	Isothermal
Wang et al., 2020	[79]	Transient	0D	Single-phase	Isothermal
Fu et al., 2021	[80]	Transient	0D	Two-phase	Isothermal
Schröder et al., 2021	[49]	Steady-state	1D (TP)	Two-phase	Isothermal
Xing et al., 2021	[81]	Transient	0D	Single-phase	Isothermal
Xu et al., 2021	[29]	Transient	1D (TP)	Two-phase	Isothermal
Gong et al., 2022	[46]	Transient	1D+1D	Two-phase	Non-isothermal
Striednig et al., 2022	[52,82]	Steady-state	0D	Single-phase	Isothermal

The cell electronic conductivity can be calculated from the series association of the electronic resistances of the different porous layers and bipolar plate [28,40,42,45,46,50,65,66,77,79] and sometimes a contact resistance [43–45,53,59,66,78] which is often a fitted parameter lumping all the electronic resistances of the cell. However, the majority of studies considered only the protonic resistance of the membrane in the ohmic overpotential. That might be a good approximation for thick membranes. But with the current trend for the use of thinner membranes, specially for mobility applications, the protonic resistances of the catalyst layers and the electronic resistances become more and more significant for the fuel cell performance. Finally, some authors [31,32,36,61,64,69] considered the ohmic resistance as a constant parameter that can be fitted to experimental data, which often correspond to normal operating conditions where the membrane is well humidified.

2.1.3. Concentration overpotential

The concentration overpotential arises due to gas transport limitations to the reaction. For detailed models where the through-plane gas concentrations are obtained, this concentration overpotential is simply accounted for by using gas concentrations in the catalyst layers for the calculation of the reversible potential and the activation overpotential. It is also possible to adopt this approach in 0D models, by calculating the gas concentrations in the catalyst layers from the channel concentrations with a quasi-steady state hypothesis for the gas transport in the porous layers, as done by some authors [39,45,52,65,66,82]. In this case, the gas concentration in the catalyst layer is given by:

$$c_{CL} = c_{GC} - \frac{i}{4F} R_T \quad (10)$$

where R_T represents a global transport resistance, which can be obtained from the effective diffusivities in the porous layers and include the transport resistance at the channel–GDL interface. This effective

diffusivity can include the effect of the liquid water saturation as in [39,66]. That allows to account for some flooding effects.

Some authors [42,43,74,87] also considered complex models to represent the local oxygen mass transport resistance in the cathode catalyst layer. While Goshtasbi et al. [43] and Futter et al. [42] used a homogeneous film model established by Hao et al. [87], Chaudhary [50] and Han et al. [74] used an agglomerate model where platinum particles form an agglomerate coated with an ionomer film to compute the local rates of reaction in the anode and the cathode CL. Both models share the need for detailed data on the catalyst layer morphology, not necessarily available. But Weber et al. [14] noted that agglomerate models, although not corresponding entirely to reality, do provide better predictions.

Most often, a specific term was used for the parametrization of the concentration overpotential [27,30–34,46,49,53,63,64,73,75,78,80,81]. It generally involves a logarithmic function of the cell current and of a limiting current, such as the one used by Schröder et al. [49] for the cathode:

$$\eta_{conc} = \left(1 + \frac{1}{\alpha_{ca}}\right) \frac{RT}{4F} \ln \left(1 - \frac{i}{i_{lim}}\right) \quad (11)$$

The limiting current i_{lim} is often a constant [30,32,34,64,78,80,81]. It can also be defined as the current for which the gas concentration is zero at the catalyst layer, which from Eq. (10) gives [88]:

$$i_{lim} = \frac{4F}{R_T} c_{GC} \quad (12)$$

This was done by a few authors such as [27,46,49,53,75].

In some cases, however, the concentration overpotential was not considered at all [35–37,51,54,55,60–62,69]. This hypothesis can stand if the fuel cell operates at low to moderate current densities.

2.1.4. Other effects

Another effect considered by Goshtasbi et al. [43] and Futter et al. [42] was the impact of platinum oxide growth dynamics in the voltage of the fuel cell. They used an ordinary differential equation to model the evolution of the platinum oxide coverage and reduce the exchange current density for the cathode overpotential accordingly. Indeed, the formation of platinum oxide occurring for high operating voltages is reported to contribute to a large extent to recoverable performance loss in PEM fuel cells [89]. Further, it should be noticed that recent experimental evidence from Bernhard et al. [90] suggests that platinum oxide growth also add ohmic losses due to increased resistance to electronic transport in the oxide layers.

Electrochemical double-layer charging and discharging is neglected by most studies, as the characteristic time scales ($\approx 10^{-6}$ s) are orders of magnitude below those of mass and heat transport phenomena. Yet, a few models considered it, like those in [1,41,42,54,61,70].

2.2. Gas transport in the porous layers

A fuel cell is composed of a number of porous layers placed between the bipolar plates: the proton-exchange membrane, the catalyst layers, the gas diffusion layers and the microporous layers. To get from the gas flow channels to the reaction sites, reactant gases must be transported across the porous GDL, MPL and CL. At the same time, water vapour produced by the electrochemical reactions at the cathode must go through those layers to reach the gas flow channels and be evacuated. In what concerns the membrane, one of its main functions is to avoid the exchange of reactant gases between the anode and the cathode. Therefore, gas transport in the porous layers plays a fundamental role in the fuel cell performance.

The anode and cathode catalyst layers are the places where electrochemical reactions occur. They combine carbon-supported catalyst nanoparticles and an ionomer binder and have a typical thickness of 1 μm –10 μm [1]. In most studies, the catalyst layer is considered as infinitesimally thin, which means that there is no accumulation of species

in it and that the conditions in the CL–DM interface prevail over all the CL thickness. Of all the studies analysed, only a few works [28,40,43,53,77] did not adopt this hypothesis. Yang et al. [28,77] considered a single control volume for the CL, which also means that there can be accumulation of species, although the transport of both across the through-plane direction in the CL was considered to be infinitely fast. Ritzberger et al. [53] did the same for the cathode CL, but considered the anode CL as infinitesimally thin. Goshtasbi et al. [43] and Zhou et al. [40] used respectively 4 and 30 control volumes to represent each CL, being able to solve transport equations in the through-plane direction across the CL. Xu et al. [29] considered one control volume for each CL for solving the transport equation for the membrane water content only, with the CL being considered infinitesimally thin for the other transport equations. Futter et al. [42] argued that in dry conditions the penetration depth of H^+ in the cathode CL is low due to low water content of the ionomer fraction, resulting in low utilization of the catalyst. Therefore, according to the authors, an appropriate model of the through-plane transport in the CL is needed in such dry conditions. To account for this phenomena with only moderate computational cost, Goshtasbi et al. [43] calculated a cathode CL utilization factor, determined from the minimization of the sum of activation and proton transport losses in the CL. It is important to note that the result of considering the CL as a single control volume or infinitesimally thin is the same if the transport equations for the different species are solved for the steady-state. The difference between both cases is the accumulation term, which is zero for the infinitesimally thin CL and non-zero for the single control volume CL, and which will have an impact only in dynamic responses.

The diffusion media of a PEM fuel cell are composed of a gas diffusion layer and in some cases of a microporous layer. The function of those layers is to ensure the proper transport of reactant gases from the gas flow channels to the catalyst layers, while ensuring the electronic and thermal conduction between the catalyst layers and the bipolar plates, the evacuation of the produced water and the mechanical support and protection of the catalyst layers. The gas diffusion layers have a typical thickness around 200 μm , while the microporous layer have a typical thickness around 50 μm . The microporous layer has a finer structure than the GDL, improving the electrical contact between the GDL and the CL and positively impacting water management [1,91]. But from a modelling point of view, the GDL and MPL are generally treated essentially in the same way. A number of studies considered the transport of gases in the diffusion media as infinitely fast [29,30,34,46,58,59,61,62,64,69,71,79,81], which is a valid hypothesis in most cases, since the characteristic time of the GDL is in the order of 1 ms–10 ms [2]. That means that the concentrations are homogeneous across the DM and are the same at the DM–CL and DM–channel interfaces. All of those works also considered the CL as being infinitesimally thin. Several authors [11,29,31,43,45,49,53,65] considered a steady-state Fickian diffusion across the DM. This allowed to account for a transport resistance caused by the DM, but without any accumulation of gas species in the DM. Others considered one [28] or more [40,47,48] control volumes across the DM while using the Fick's law, therefore allowing for accumulation effects. In the case of Ziogou et al. [33], steady-state Fickian diffusion was considered for oxygen and hydrogen, but water vapour can accumulate in the DM control volume. Several authors [42,44,55,68] used the more complex Stefan–Maxwell diffusion. The work of Lindstrom and Wetton [92] indicated that Fickian diffusion can be a good approximation for the gas transport in the cathode DM when air is used as oxidant in the fuel cell. However, the same authors showed that this approximation is not good for the anode DM if hydrogen and nitrogen are mixed. Such a condition can be found with anode gas recirculation or in dead-end anode operation, where nitrogen crossing over from the cathode can accumulate in the anode [72,93]. Only Chaudhary et al. [50] and Futter et al. [42] considered the hydraulic permeation of the gas phase in the through-plane direction across the DM, which was modelled according to Darcy's

law. Most authors neglected this hydraulic permeation arguing that the gas pressure gradient is negligible across the porous layers. Qin et al. [70] considered hydraulic permeation in the in-plane direction for serpentine flow-fields by calculating the share of the flow going through the DM with 3D CFD simulations as was done in the work of Park and Li [94]. The case of water vapour is particular, as it may be coupled to liquid transport through phase-change. Zhou et al. [40] for instance only solved the transport equation for water vapour and not for other gases. In their steady-state model, Jiang et al. [11] compared the net water transfer flux across the membrane to the evaporation flux at the DM-channel interface to determine whether the water transport through the DM happens in the liquid or vapour form, in which case the Fick's law is applied.

It is also important to note that although several authors ignore gas transport phenomena across the diffusion media and the CL, its impact on the cell voltage is most often still considered through a parametrization of the transport overpotential as explained in Section 2.1.

The effective diffusivity in porous media such as the GDL or the CL is obtained by multiplying the bulk diffusivity by a correction factor to account for porosity and tortuosity effects. Most often, this factor consists in a power function of the medium porosity such as:

$$D^{eff} = D^{bulk} \epsilon^{1.5} \quad (13)$$

or :

$$D^{eff} = D^{bulk} \left(\frac{\epsilon - \epsilon_p}{1 - \epsilon_p} \right)^{0.785} \quad (14)$$

A discussion of the different expressions can be found in the review paper of Andersson et al. [8]. They conclude that some expressions which are used in the fuel cell modelling community may not be valid as they were obtained for porous media with different characteristics from those of fuel cell diffusion media, leading to an overestimation of the effective diffusion coefficients. Corrections to account for the saturation of the medium can also be applied if liquid water is present. Those corrections are discussed in Section 2.3. Knudsen diffusion may also be considered, as it can be significant for the small pore sizes of the CL and the MPL [2,14]. Knudsen diffusion was included in the works of Goshtasbi et al. [43] and Futter et al. [42], for instance.

Concerning the proton-exchange membrane, the permeation of species other than water, namely N_2 , O_2 and H_2 , was most often neglected. Yang et al. [28,77] considered the crossover of nitrogen only, with a correction factor for the crossover rate to match experimental data. Ritzberger et al. [53] also considered only the crossover of nitrogen. Differently, Futter et al. [42] considered the transport of O_2 and H_2 across the membrane. Among the models discussed in this work, the ones of Goshtasbi et al. [43], Tang et al. [41] and Chen et al. [78] are the only that took into account the crossover of N_2 , O_2 and H_2 . The transport of water across the proton-exchange membrane is treated specifically in the Section 2.4 of this work.

2.3. Liquid water and phase-change in the porous layers

The consideration of liquid water and phase-change is perhaps one of the most critical aspects for water management. Through-plane transport of liquid water in the porous layers is a complex issue and several approaches to handle (or avoid) it can be found in the literature. Some authors such as Striednig et al. [52] and Abdin et al. [65] considered that only water vapour exists in the stack. Liso et al. [68] considered that water in the liquid phase is finely dispersed in the gas phase, and no distinction is made between the two phases. The authors recognized that this can only be a valid assumption if the amount of excess liquid water is low. This is also the case for some other authors [59,62,71] although this hypothesis was not so explicit. In some works [27,30,58] the cathode and the anode sides were modelled as a single control volume each (infinitesimally thin GDL, MPL and CL) and therefore there was no liquid transport in the through-plane direction,

even though liquid water could accumulate in those control volumes. Damour et al. [61] did not consider the presence of liquid water, considering only the gas streams to be saturated with water vapour in order to compute the membrane conductivity. The same hypothesis was adopted by Schultze and Horn [69]. Ritzberger et al. [53] considered that although liquid water could exist in a control volume (the CL or the channel), water transport between two control volumes could only happen under the gas phase. De Campo [73] considered that the water could accumulate in the gas channels through condensation, but could be removed by convection with the same velocity as the gas phase. Wang and Chen [95] proposed a non-dimensional Damköhler number defined as the ratio between water production at the CL and water removed by vapour diffusion across the diffusion media to characterize the existence of two-phase flow in the diffusion media. A Damköhler number close to zero indicates single-phase operation, while a value much greater than unity indicates two-phase conditions everywhere in the cathode. This non-dimensional number can then be used to evaluate the need for a two-phase model where liquid water transport is considered.

Liquid water transport is most often modelled using Darcy's law [11, 28,29,31,36,40,43,45–50]. The liquid water flux is then written:

$$q_l = - \frac{\rho_l K_l^{eff}}{M_{H_2O} \mu_l} \frac{dp_c}{dx} \quad (15)$$

with the effective permeability calculated as $K_l^{eff} = K_{l,abs} K_{l,rel}$. The relative permeability $K_{l,rel}$ is modelled as a function of the water (and ice) saturation : $K_{l,rel} = s^n$ or $K_{l,rel} = s^n (1 - s_{ice})^n$. The exponent n can take prescribed values such as 5 [41,43], 4 [11,40,57], 3 [28,29,31,36, 39,45,46,50], 2.5 [42] or even be fitted during model calibration [49]. Chaudhary et al. [50] added to Eq. (15) a term to account for the shear force exerted by the gas-phase velocity over liquid-phase, as they also considered the hydraulic permeation of the gas phase.

A closure relation between the capillary pressure and the water saturation completes Eq. (15):

$$p_c = \frac{\sigma_l \cos \theta_l}{(\epsilon_l K_l)^{1/2}} J(s) \quad (16)$$

$$\frac{dp_c}{dx} = \frac{dp_c}{ds} \frac{ds}{dx} = \frac{\sigma_l \cos \theta_l}{(\epsilon_l K_l)^{1/2}} \frac{dJ}{ds} \frac{ds}{dx} \quad (17)$$

The most commonly adopted parametrization for the so-called Leverett J-function $J(s)$ has been obtained by Udell [96] by fitting experimental data for sands from Leverett [97]:

$$J(s) = 1.417s - 2.120s^2 - 1.263s^3 \quad (18)$$

It was used by many authors [11,28,29,31,36,39,40,42,43,46,57] although its applicability has been questioned [1,8] as it has been obtained for the water transport in soil materials. Alternatively, Schröder et al. [49] used a fitted function for a SGL10BA GDL taken from Gostick et al. [98] and Chaudhary et al. [50] a fitted function obtained by Kumbur et al. [99] for the SGL series. However, Jiao and Li [57] and more recently Vetter and Schumacher [12,13] found that the J-function parametrization had little impact on the simulation results.

In the previous Eqs. (15)–(18), some authors [31,49,66] used, instead of the actual saturation s , a reduced saturation defined as:

$$s_r = \begin{cases} \frac{s - s_{im}}{1 - s_{im}} & \text{if } 0 < s \leq 1 \\ 0 & \text{if } s = 0 \end{cases} \quad (19)$$

where s_{im} is the immobile water saturation level. s_{im} was considered equal to 0.08 in Schröder et al. [49]. According to Gerteisen et al. [56], immobile water saturation correspond to liquid water located in hydrophilic regions of the GDL. These authors assumed $s_{im} = 0.2$ in the GDL and zero in the CL, a value also adopted by Goshtasbi et al. [66]. Gostick et al. [98] measured values between 0.04 and 0.13, depending on the GDL material.

The liquid water flux being calculated with Eq. (15)–(18), conservation equations can be written in several forms, especially depending on how the control volumes are discretized. In del Real et al. [31] the liquid water was considered to occupy uniformly the diffusion medium volume (with a infinitesimally thin CL). The saturation is considered zero at the channel so that the liquid water flux was defined as :

$$q_l = -\frac{\rho_l K_l^{eff}}{M_{H_2O} \mu_l} \frac{dp_c}{ds} \frac{s}{t_{GDL}} \quad (20)$$

A similar hypothesis was adopted by Strahl et al. [36]. We must note that $t_{GDL}/2$ should be used instead of t_{GDL} in Eq. (20) to be consistent with the cell-centred discretization used. Further, produced water was considered as a water vapour source in vapour form in the GDL, with the possibility of evaporation/condensation.

Yang et al. [28,77], instead, used one control volume for each porous layer (CL, MPL, GDL) and a careful treatment of effective transport properties between two layers. Further they solved a partial differential equation for the liquid pressure, rather than for the liquid saturation. The liquid pressure continuity across the interface between two layers justified this choice. The authors only used a water saturation conservation for the gas channel control volume, considering convective liquid water transport. This convective transport was calculated from the liquid velocity, obtained by applying a slip ratio to the gas velocity. In [77] the liquid velocity was considered to be the same as the water vapour velocity, although this hypothesis was not explicit in [28]. Evaporation/condensation were also considered as source-terms in the transport equations both in the porous layers and the gas channel. A similar approach was adopted by Tang et al. [41, 100]. But these authors used a modification of the Lockhart–Martinelli correlation to compute the liquid–gas slip ratio, instead of assuming it equal to unity. Gong et al. [46] solved the same equations as Yang et al. [28] but instead of one control volume for each porous layer, they used a single control volume to represent the combination of CL, MPL and GDL and considered the water saturation to be homogeneous across those layers.

Salva et al. [45] considered one control volume for each layer (CL, MPL, GDL) as in Yang et al. [28,77], but solved the transport equation for the steady-state and assume equilibrium between liquid and vapour phases in each layer, with infinitely fast evaporation and condensation. While the infinitely fast evaporation corresponding to a local equilibrium of liquid and vapour phases was assumed by a number of authors, it can bring numerical difficulties due to the discontinuity arising from water condensation. Strategies to smooth the transition between unsaturated and supersaturated conditions were proposed by Chen et al. [78] and Ritzberger et al. [53].

Kang [47] solved a partial differential equation for the total water transport in the GDL, which was discretized with four control-volumes in the through-plane direction. Evaporation and condensation were assumed to happen infinitely fast and the resulting saturation was used to calculate a liquid water flux between two adjacent control volumes, included in the total water transport equation. The CL was supposed to be infinitesimally thin. The authors said that the presence of liquid water in the gas channels is neglected, but it was not clear how this assumption was considered in the model equations.

Schröder et al. [49] proposed an original approach. The authors solved the mass conservation equation to obtain the total amount of water accumulated in the reference volume (gas channel + GDL pores). From that, the authors calculated the relative humidity in the reference volume. The water flow from the CL to the gas channel was considered to be equal to the sum of water produced from the reaction and transported across the membrane. If the relative humidity in the reference volume was equal to 100 %, all the water flow from the CL to the gas channel was considered to be in liquid form. The authors then solved a 1D steady-state transport equation for liquid water in order to obtain the reduced water saturation profile $s_r(x)$ across the GDL. An average reduced saturation over the GDL thickness was then obtained.

This average value was finally scaled with a fitted parameter to mimic along-the-channel effects.

Goshtasbi et al. [43] and Zhou et al. [40] solved the 1D partial differential equation with more than one control volume for each porous layer. Goshtasbi et al. used 20 for each GDL, 4 for each MPL and 4 for each CL, but vapour and liquid equations were decoupled. First the vapour concentrations were obtained through a quasi-steady approximation. These vapour concentrations allowed for the calculation of phase-change source terms for the transient liquid transport equation. Vapour fluxes were updated to account for phase-change. An average saturation in the GDL was used to modify the gas diffusivities in turn. Readers should refer to [66] for more details on this method. Zhou et al. used 30 for each GDL and 20 for each CL and also decouples the liquid transport from the vapour transport, which was solved with a single control volume for the GDL and a single control volume for the CL.

Finally, Xu et al. [29] used analytical solutions for the transport of liquid water across the GDL obtained by Hu et al. [101] from the integration of 1D through-plane transport equations for three different regimes: unsaturated vapour throughout the GDL, saturated vapour throughout the GDL and a mixed regime where a saturation front exists. A similar approach was adopted by Jiang et al. [11], where a switching function was used to determine whether the water was transported as liquid or vapour in the porous layers.

It should be noted that many papers use as boundary condition between the GDL and the gas channel a fixed small (or zero) saturation, since any liquid water would be quickly removed by the gas stream [11, 33,40,50]. The exceptions are works of Jiao's group [28,46,77] where a saturation conservation equation was solved for the channel, Gerteisen et al. [56] where the authors imposed a liquid water flux proportional to the gas flow rate and the square of the difference between the saturation s and its immobile part $(s - s_{im})^2$. On the membrane–CL interface, most papers used a zero-flux boundary condition. Exceptions are the works of Tang et al. [41,100] that considered the possibility of hydraulic permeation of liquid water from the CL into the ionomer and the one of Chaudhary et al. [50] that considered separate liquid and vapour water absorption/desorption by the membrane.

The main effect of the presence of liquid water (or ice) in the electrodes and porous layers is the blockage of the reactant's access to the catalytic sites. It can be taken into account in different ways. Many studies [11,28,29,39,40,42,43,45,46,49,50,57,66,72,74,76,77] used an effective diffusivity to calculate gas transport which includes the effect of the liquid water saturation in the wet porous layers:

$$D_{k,wet}^{eff} = (1 - s)^{n_s} D_{k,dry}^{eff} \quad (21)$$

with n_s usually varying between 1 and 3 [2,15]. The ice volume fraction can be added to the liquid saturation when pertinent. This reduction in the effective diffusivity will increase the gas transport resistance and thus impact the concentration overpotential of the cell (see Section 2.1.3). In some works [42,43,56,74] where agglomerate or film models were used for the catalyst layer, the liquid saturation in the CL add an oxygen transport resistance due to the presence of a liquid film covering the ionomer film. Other authors [31,36,40,41,47,48] considered the effect of liquid water through a reduction in the effective cell active area proportional to the water saturation:

$$A_{cell}^{eff} = (1 - s) A_{cell} \quad (22)$$

Strahl et al. [36] adopted an original approach by considering that the water saturation of the cathode catalyst layer increase the active area until an optimum saturation value is attained, after which the active area remain constant:

$$A_{cell}^{eff} = A_{opt} [\exp(k_s s) - 1], \quad 0 < s < s_{opt} \quad (23)$$

A_{opt} and k_s are both fitting parameters. Strahl and Costa-Castelló [102] proposed the following equation in a latter work:

$$A_{cell}^{eff} = A_{opt} \left(1 - \frac{s_{opt} - s}{s_{opt}} \right)^{1/3} \quad (24)$$

with the fitting parameter s_{opt} . As we will see in Section 4, adopting this relation between active area and liquid saturation allowed the model to better fit transient voltage data. The increase of the active area with liquid saturation is in line with experimental observations reported by Padgett et al. [103], where platinum utilization increased with the relative humidity for porous support materials. The authors argued that condensed water allows the utilization of platinum catalyst present in pores that are too small for the ionomer to be in. Finally, it is important to note that the works of Strahl et al. [36,102] concerned open-cathode fuel cells where the cathode stoichiometry is rather high (>20).

2.4. Membrane water transport

Water transport across the PFSA membrane is essential to water management, as it will define water content and humidity levels across the MEA. While readers should refer to the work of Kusoglu and Weber [104] for a comprehensive review on physical phenomena occurring in the membrane, the review of Dickinson and Smith [105] presents very well approaches for the continuum modelling of water fluxes across the membrane and the CL ionomer. Most often, system-level studies adopt the Springer membrane model [84] where the water flux in the membrane is considered as the sum of a Fickian diffusion term and electro-osmotic drag term [105]:

$$q_w = -D_\lambda c_f \nabla \lambda + \xi \frac{i}{F} \quad (25)$$

The parameters D_λ and ξ depend on the material as well as on the temperature and the membrane water content, with several parameterizations available in the literature. An alternative approach is to use the model proposed by Weber and Newman [86,106], as Futter et al. [42] did. Sorption isotherms are also available and give the equilibrium water content λ_{eq} as a function of the temperature and the water activity. These sorption isotherms have generally been obtained for Nafion 1100 and Dickinson and Smith [105] cautioned against their use for other materials even in the Nafion family. Such use is nevertheless a common practice due to a lack of data for specific materials. Authors such as Goshtasbi et al. [43] and Del Real et al. [31] tried to go around the problem by fitting a correction factor for the water diffusion coefficient, while Abdin et al. [65] used a constant water diffusivity coefficient. The reader should refer to the works of Dickinson and Smith [105] and Vetter and Schumacher [12] for discussions and comparisons of the different parameterizations available for the membrane water diffusivity, electro-osmotic drag coefficient and sorption isotherms.

System-level studies have proposed different approaches to calculate the water flux across the membrane and the membrane water content. The simplest one is to consider that the membrane has a constant water content, often corresponding to well-humidified conditions [61,64,69,73,79,81]. Strahl et al. [36] went further and neglected the water flux across the membrane, arguing that electro-osmotic and diffusive fluxes have the same order of magnitude and therefore cancel each other for the low current densities where their open-cathode fuel cell stack operates. A popular alternative consists in considering that the membrane-CL interfaces are in equilibrium with water contents λ_{ca} and λ_{an} (calculated from the water activities in each CL) and to consider the membrane water content to be the average value as in the work of Prukushpan et al. [30]:

$$\lambda_{memb} = (\lambda_{ca} + \lambda_{an})/2 \quad (26)$$

with the gradient $\nabla \lambda$ calculated as:

$$\nabla \lambda \approx (\lambda_{ca} - \lambda_{an})/t_{memb} \quad (27)$$

Similarly, some authors [49,62,63] adopted a linear water content profile across the membrane:

$$\lambda(x) \approx \lambda_{an} + x \times (\lambda_{ca} - \lambda_{an})/t_{memb} \quad (28)$$

and use it to obtain average values of D_λ and ξ . Del Real et al. [31] and Ziogou et al. [33] used instead the anode water content to calculate the diffusivity across the membrane and the electro-osmotic coefficient. Oppositely, Huisseune et al. [37] considered the membrane water content as being constant and equal to the cathode water content λ_{ca} . A more common approach [34,45,60,73] is to consider a membrane water content corresponding to the equilibrium water content for an average water activity given by:

$$\lambda_{memb} = \lambda_{eq} \left(\frac{a_{ca} + a_{an}}{2} \right) \quad (29)$$

These approaches do not consider the possibility of water accumulation in the membrane and no conservation equation is solved for a membrane control volume in the membrane. That means that they cannot reproduce the dynamics of membrane sorption.

In an attempt to describe the dynamics of water accumulation in the membrane, Ritzberger et al. [53] considered the membrane water activity to follow the average $(a_{ca} + a_{an})/2$ as in the work of Salva et al., but with a time constant τ_{memb} :

$$\dot{\lambda}_{memb} = -\frac{1}{\tau_{memb}} \left(\lambda_{memb} + \frac{a_{ca} + a_{an}}{2} \right) \quad (30)$$

Goshtasbi et al. [43,66] considered hydraulic liquid water and thermo-osmotic transport in addition to electro-osmotic transport and Fickian diffusion, yielding:

$$q_w = -D_\lambda c_f \nabla \lambda + \xi \frac{i}{F} - \frac{K_l}{\mu_l M_{H_2O}} \nabla p_l + D_T \nabla T \quad (31)$$

Hydraulic permeation was also accounted for in a few other works [11, 37,75]. Dickinson and Smith [105] mentioned that hydraulic permeation might play a role only for liquid-equilibrated conditions.

In terms of membrane water conservation, Goshtasbi et al. [43,66] solved a conservation equation for three control volumes corresponding to the whole membrane thickness, and for the ionomer in each of the catalyst layers. This was also done by Yang et al. [28,77] and Xu et al. [29]. Kang [47] considered a single control volume for the membrane thickness, but no conservation equation for the water content in the CL ionomer, with diffusion fluxes calculated assuming equilibrium water contents in the membrane-CL interfaces. Zhou et al. [40], however, considered 50 control volume for the through-plane discretization of the membrane.

Only some authors [28,29,40,43,68,77] considered the non-equilibrium dynamics of water absorption (or desorption) into the ionomer, with the absorption flux q_{abs} written:

$$q_{abs} = k_{abs} c_f (\lambda_{eq} - \lambda) \quad (32)$$

In particular, Liso et al. [68] showed with steady-state water balance measurements that considering such non-equilibrium conditions lead to better prediction of membrane water fluxes, while considering equilibrium water content at the membrane-CL interfaces yielded over-estimated membrane water fluxes. They used a Nafion 112 membrane. Majsztrik et al. [107] also provided evidence of the importance of the interfacial transport resistance for the membrane sorption, specially for thin membranes. Xu et al. [29] adopted an weighted average of the absorption/desorption rate constants for vapour and liquid water to account for faster kinetics when the membrane is in contact with liquid water, with the weighting factor depending on both the ionomer water content and the water activity. Dickinson and Smith [105] recommended the use of the water activity difference between the membrane and the gas phase as the driving force, rather than the membrane water content as in Eq. (32):

$$q_{abs} = k_{abs} (a_{ca/an} - a_{memb}) \quad (33)$$

Further, they mentioned that there is no consensus on whether an active electro-osmotic water uptake should be considered, yielding:

$$q_{abs} = -\xi \frac{i}{F} + k_{abs} (a_{ca} - a_{memb}) \quad (34)$$

$$q_{abs} = \xi \frac{i}{F} + k_{abs}(a_{an} - a_{memb}) \quad (35)$$

One aspect that is often neglected in the modelling of membrane water absorption/desorption dynamics is the so-called Schröder's paradox, consisting in different membrane water uptakes depending on if the membrane is in contact with liquid water or saturated water vapour [2, 105]. Chaudhary et al. [50] evaluated two approaches to account for Schröder's paradox, both considering individual contributions for membrane water absorption/desorption from vapour and liquid phases, with membrane water desorption occurring only in the liquid phase. In a first approach, different membrane water equilibrium concentrations were considered depending on the water phase, liquid or vapour. In the second approach, a single value of the membrane water equilibrium concentration was considered for both phases. The authors showed that the first approach gives better results when the anode and cathode inlet gases are fully humidified (RH = 100 %), which is probably linked to the presence of liquid water. Futter et al. [42] accounted for the Schröder's paradox by using in the membrane transport equations an expression for the chemical potential of water depending on the local water saturation.

Another aspect that many authors neglected is the fact that ionomer thin-films in the catalyst layer might behave significantly differently from the bulk material in the membrane, as deeply discussed by Weber et al. [14]. Differences include lower equilibrium water content and conductivity or higher activation energy for the thin-films as compared to the bulk ionomer. It is remarkable that no study took these effects into account.

2.5. Heat transfer

In fuel cells, heat is released by electrochemical reactions and ohmic resistances. Water condensation or evaporation constitute a heat source as well. Most heat is released in the MEA, due to entropic and irreversible reaction heat in the CL and protonic resistance in the membrane and the ionomer phase of the CL [108]. In liquid-cooled or air-cooled fuel cells, the excess heat is evacuated by the coolant flowing in channels in the bipolar plates. Due to water phase-change and thermal gradients, water and heat transport are tightly coupled, so that heat transfer is key for both water and thermal management.

The majority of system-level studies considered a homogeneous stack temperature, identical in all cells, across all the cell layers and across all the cell surface [30–32,34,36,51,53,58,59,61,62,64,65,68,69, 71,73,79–81]. In their work, Schröder et al. [49] also considered a homogeneous stack temperature, but with a different temperature for the gas in the cathode gas flow channels. In some works as the ones of Hoeflinger and Hofmann [51] or Abidin et al. [65], the temperature was an input parameter of the model and considered to be constant. Most often [32,34,36,73,81], however, a single energy balance equation was solved, with the general form:

$$C_{st} \frac{dT_{st}}{dt} = \sum (\dot{m}h)_{in} - \sum (\dot{m}h)_{out} - P_{el} - \dot{Q}_{cool} - \dot{Q}_{amb} \quad (36)$$

It should be noted that this hypothesis is less valid in some cases. For short stacks with few cells, heat losses through the end plates might be significant so that the cells are not the same temperature. For high surface area cells, temperature differences across the cell surface might be of importance. This is specially the case for high current densities, where the amount of heat released is high and thermal gradients are strong. Nandjou et al. [10] put in evidence temperature differences of up to 10 °C at 0.8 A cm⁻² for a cell with an active area of 220 cm². Wang et al. [109] also indicated that temperature differences across the GDL can reach 5 °C–10 °C. Due to the steep relation between temperature and saturation pressure of water, small changes of temperature may lead to strong variations of relative humidity. For instance, 100 % RH at 75 °C gives only 81 % RH at 80 °C or 67 % RH at 85 °C if the water vapour pressure is kept constant. This highlights the fact that only

small changes in temperature can have significant impact for water management.

Tang et al. [41,100] treated the solid and gas with different temperatures in the CL and the GDL, although the fluid and solid domains were supposed to be in thermal equilibrium in the membrane. In the CL and the GDL, the condensing liquid film was assumed to be at the same temperature as the solid.

Salva et al. [45] on the other hand, considered the gas-phase to be in equilibrium with the solid layer they are flowing in, each of these layers (membrane, CL, MPL, GDL, BPL) having an homogeneous temperature. Heat could be transferred between the layers by conduction, characterized by heat transfer resistances. An originality of the model is that each layer can lose heat to the environment at its perimeter. Reaction heat was considered to be fully released in the cathode CL. The authors didn't mention any kind of active cooling, but it could be easily accounted for in the equations, as was done in Yang et al. [28]. Instead, they considered the temperature of the cathode BP as known and used it as an input parameter. The equation for the cathode GDL heat balance, for instance, was written:

$$\frac{T_{GDL} - T_{CL}}{R_{CL-GDL}} + \frac{T_{GDL} - T_{BP}}{R_{GDL-BP}} + A_{lat,GDL} h(T_{GDL} - T_{amb}) = 0 \quad (37)$$

The authors' approach implies that heat capacities of gases and liquid water inside the cell is negligible, and that equations are solved in steady-state. Further, they do not seem to account for the latent heat released by water condensation in the energy balances.

Yang et al. [28,77] adopted an approach similar to Salva et al. for the through-plane discretization of the heat equation, but considered transient effects with a forward Euler discretization. For the cathode GDL, it gives:

$$T_{GDL}^t = T_{GDL}^{t-\Delta t} + \frac{\Delta t}{(\rho C_p)_{GDL}} \left(\frac{T_{GDL} - T_{CL}}{R_{CL-GDL}} + \frac{T_{GDL} - T_{BP}}{R_{GDL-BP}} + S_T \right) \quad (38)$$

where S_T is the heat source term which included different terms depending on the layer. The thermal resistances R_{i-j} between layers i and j were calculated by using effective thermal conductivity values which account for the fractions of the different phases in each layer (gas, liquid and solid) [77]. Heat could be also exchanged between different cells of the stack, which were considered individually. The same approach was adopted by Zhou et al. [40] or Chaudhary et al. [50], although these authors used a more detailed discretization in the through-plane direction.

Gosthasbi et al. [43] and Kang [47] also adopted this approach but with some differences. Gosthasbi et al. [43] took the gas channel temperatures as boundary condition. The authors supposed the gas channel temperature in the k th channel segment to be equal to the cooling fluid temperature in that segment, which in turn was given by a linear dependence on the local current density i_k :

$$T_{cool,k} = T_{cool,in} + \frac{\sum_{j=1}^k i_j}{\sum_{j=1}^{N_{seg}} i_j} \Delta T_{cool} \quad (39)$$

Kang [47] considers also convection heat transfer in the gas diffusion layer.

It should be noted that a recent numerical study by Sauermoser et al. [110] indicates that Peltier and Dufour effects impact significantly the heat fluxes in the cell. As far as we know, these effects have not been taken into account by any system-level study so far.

2.6. Flow in the gas channels

Most studies considered a single control volume to represent the gas flow channel [30,31,34,53,59–61,71,73,81], which is perhaps the simplest approach. In this case, the mass conservation equation for each gas species is written in the general form:

$$\frac{dm_{k,GC}}{dt} = \dot{m}_{k,in} - \dot{m}_{k,out} + S_k \quad (40)$$

This equation was often combined with linear nozzle equations to determine the inlet and outlet mass flow rates from the difference of pressure between the gas flow channel control volume and upstream or downstream pressures [30,53,71]. The mass source term S_k can include any transport of species from the channel to the GDL. Schultze and Horn [69] adopted a similar model but used it to derive a first-order transfer function to represent the cathode gas flow channel. The time constant of this transfer function was fitted to experimental data.

Other authors used a one-dimensional discretization along the channel direction to capture heterogeneity over the cell surface. Goshtasbi et al. [43] used 12 or 27 control volumes along the channels, depending on the channel length, as the authors noticed that control volume lengths of more than 1 cm lead to numerical issues. The authors considered that no species can accumulate in the channel so that for each segment i the following equality holds:

$$\dot{m}_{k,out}^i = \dot{m}_{k,in}^i + S_k^i \quad (41)$$

Further, the authors used an iterative procedure to simulate counter-flow configurations. Yang et al. [28] adopted a similar approach to the one of Goshtasbi et al. [43], combining an explicit time discretization and a first-order upwind discretization along the channel, so that the total mass flow at the outlet of a channel segment could be calculated from the inlet mass flow and other mass sources at the previous time step:

$$\dot{m}_{out}^{i,t+\Delta t} = \dot{m}_{in}^{i,t} - \sum_k S_k^{i,t} \quad (42)$$

This approach is explained in Wang et al. [72]. The use of an explicit time discretization allowed the consideration of counter-flow configurations, but at the expense of small time steps (of the order of 10^{-6} s). Interestingly, Yang et al. [28] gave the results of a convergence study showing that the three nodes along the channel are sufficient to keep the relative error within $\pm 0.5\%$ as compared to ten nodes for the outlet gas velocity, the water saturation in the CL and the output power of the stack. With a single node, the relative error remained within $\pm 1\%$ for the two latter parameters, but attained near 6% for the outlet gas velocity. Kang [47] adopted a 2D discretization along the channel with 20 control volumes, but only convection in the flow direction and through-plane transport in the porous layers were considered. The only 2D transport considered was the heat conduction in the bipolar plates. Pressure drop along the channels was calculated from the friction factor, but it is not clear how outlet flow rates for each control volume were calculated.

Schröder et al. [49] adopted a pragmatic approach to deal with the problem of along-the-channel heterogeneity. The total volume of fluid leaving each gas channel was calculated for steady-state conditions, while water was allowed to dynamically accumulate in the cathode or the anode. The cathode oxygen mole fraction and the cathode pressure were taken as the average between the inlet and outlet values. However, the cathode fluid temperature was calculated from a fitted linear combination of inlet and outlet temperatures. An effective relative humidity was also calculated by multiplying the cathode relative humidity by a fitted coefficient. The authors argued that this allows to take into account the non-linear dependence of the ionomer's ohmic resistance on the relative humidity. It should be noted that some authors such as Lazar et al. [75] used a logarithmic average to account for non-homogeneous distribution of concentrations along the gas flow channel, but without any justification.

Falcão et al. [38] adopted another approach by considering that the concentrations of gaseous species at the channel outlet were equal to the concentrations at the GDL-channel interface, and that the gas volume flow rate was constant along the channel. Their model assuming steady-state operation, balance equations could be written such as the one for the oxygen on the cathode side:

$$\frac{i}{4F} = \frac{\dot{Q}_{ca,in}}{N_{cells}A_{cell}} \left(c_{O_2,GC}^{in} - c_{O_2,GC-GDL} \right) \quad (43)$$

If all the other parameters are known, that allows one to obtain the boundary condition of the through-plane transport, $c_{O_2,GC-GDL}$. Salva et al. [45] also adopted this approach.

One important aspect in the modelling of the gas flow channels is the liquid water transport along the channel. Several authors [31,34,48,53,58,64,69] considered that only water vapour was transported along the channel and could leave the stack. In particular, Zhao et al. [64] considered that the cathode and anode outlet gases were fully saturated ($RH = 100\%$) with no liquid water. Schultze and Horn [69] also adopted this hypothesis for the cathode, but did not include the anode channel in their model. Conversely, Qin et al. [70] supposed that the anode outlet gas is fully saturated with no liquid water, instead of the cathode. Zhou et al. [40] and Strahl et al. [36] considered that the water saturation is zero in the channel arguing that any liquid water is quickly removed by the gas flow. Del Real et al. [31] allowed for liquid water removal during anode purge events, where they considered that all liquid water was removed from the anode. Some authors [43,49,62,68] assumed that liquid water was present in a dispersed phase in the gas flow channels and used a total water molar flow in the calculations, representing the sum of liquid and vapour water in the channel. Yang et al. [28] used a fixed velocity ratio to obtain the liquid phase velocity from the gas velocity in the channels. The value of the velocity ratio was not mentioned, but in a previous work of the same group [72] this velocity ratio was equal to one, indicating that a dispersed phase was also assumed. A velocity ratio equal to one was also adopted by De Campo [73], while Tang et al. [41,100] used a modification of the Lockhart-Martinelli correlation to compute the liquid-gas velocity ratio. In the analytical solutions for the through-plane water transport used in the model of Xu et al. [29], an interfacial drag assumption was adopted, with the liquid velocity given by:

$$v_l = v_g \frac{s_{GC}^3}{(1 - s_{GC})^3} \frac{\mu_g}{\mu_l} \quad (44)$$

Alternatively, Schultze and Horn [60] and Fly [63] considered the mass flow rate of liquid water leaving the cathode to be proportional to the gas mass flow rate. While Schultze and Horn did not mention which value they used as a proportionality constant, Fly fixed an arbitrary value of 2.

Finally, it is important to mention that a few authors [11,28,39,43,48] considered a mass transfer resistance between the channel gas flow and the GDL by using Sherwood number correlations, usually valid for fully developed laminar flow, but most neglected it. For two-phase conditions where droplets or films might be present in the channel, Sherwood number correlations were provided by Koz and Kandlikar [111–113].

2.7. Cell and stack heterogeneity

Operating conditions are not homogeneous over the active area of a fuel cell, as evidenced by the work of Miao et al. [114]. These conditions can also differ between different cells of a stack, with for instance temperature variations across the stack due to heat losses at the end plates. However, modelling those heterogeneous conditions require a level of spatial detail that most authors choose to neglect. Some attempts to consider them in a computationally efficient way have been made, nevertheless.

A first approach to consider heterogeneous conditions over the cell surface area is to use some kind of 1D+1D model, where in addition to the solution of the through-plane 1D transport equations, a 1D discretization along the channel direction is adopted, as described in Section 2.6. Kang [48] was even able to account for some 2D effects over the cell active area for a serpentine flow field with their quasi-3D modelling. A serpentine flow field can also be modelled by the flow network model proposed by Qin et al. [70]. But discretization along the channel is not enough to take into account the effect of the bipolar plate land area, where the plate is in contact with the GDL.

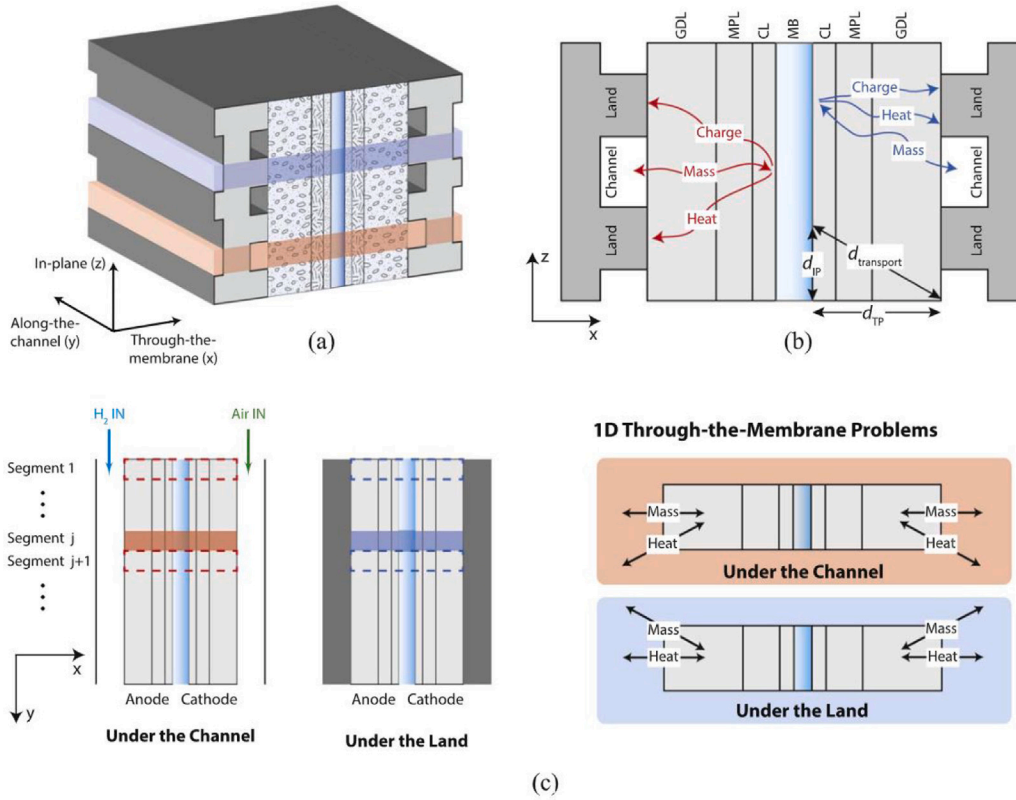


Fig. 3. Schematics of the bi-domain approach used by Goshtasbi et al. [43] to represent channel-land heterogeneity: (a) under the channels (red) and under the land (blue) domains, (b) transport paths considered for these different domains, (c) representation of the 1D+1D approach decoupling along-the-channel and through-plane directions. Source: Reproduced from Goshtasbi et al. [43].

Goshtasbi et al. [43] adopted a bi-domain approach, initially proposed by Zaglio et al. [115]. This approach is based on solving the through plane equations for two different domains: one under the channel and one under the land, with respective effective transport lengths and boundary conditions. It is schematized in Fig. 3. The effective transport length takes into account geometrical lengths but also anisotropy in the transport properties of materials. Schröder et al. [49] and Tang et al. [41] used a conformal mapping to calculate an effective length for each layer, as proposed by Weber [116]. In a simpler approach, Yang et al. [28] took the channel-land heterogeneity by considering only the channel area in the channel-GDL interface. None of the other works reviewed here considered channel-land heterogeneity.

In what concerns stack heterogeneity, all works except the ones of Zhou et al. [40] and Yang et al. [28] considered all the cells to operate identically, with the same conditions. Being so, the solution for a single cell was obtained and the resulting voltage and flow rates are multiplied by the number of cells in the stack to obtain results for the stack. Zhou et al. and later Yang et al. considered instead each cell individually, with heat exchange between adjacent cells and heat losses to the environment for the two end cells.

3. Modelling auxiliary components

The fuel cell stack model is at the heart of any fuel cell system modelling effort. However, having appropriate models of the different auxiliary components in the cathode, anode and cooling loops is almost as important for system-level studies. Indeed, they allow to determine the operating conditions to which the stack is subject, specially in case of variable load or environmental conditions. For instance, the fuel cell stack model may require as an input the coolant temperature at the inlet of the stack's cooling channels. This temperature is provided as an output of the cooling loop model. But in turn, the cooling loop

model requires the coolant temperature at the outlet of the stack as an input, which is calculated by the fuel cell stack model. Being so, when integrated in a system-level model, the models of the fuel cell stack and the auxiliary components are often interconnected. The models of the auxiliary components also allow to estimate the system efficiency, for instance by considering the power consumed by the air compressor, or to evaluate the water and heat balances at the system level. In this section we review the modelling approaches adopted in the fuel cell modelling literature to account for some important components in fuel cell system studies.

3.1. Air compressor

The air compressor is a key element of the fuel cell system as it allows to increase the air pressure in the cathode, thereby improving the efficiency of the fuel cell stack. Several authors [28,30,55,59,71,73,74,80] modelled the compressor by considering its dynamics, including a momentum conservation equation for the motor shaft and an equation for the DC motor torque depending on the applied voltage. For instance, Pukrushpan et al. [30] used the following equation to determine the compressor rotational speed ω_{cp} :

$$\frac{d\omega_{cp}}{dt} = \frac{\tau_{cm} - \tau_{cp}}{J_{cp}} \quad (45)$$

The compressor motor torque τ_{cm} is given by:

$$\tau_{cm} = \eta_{cm} \frac{k_t}{R_{cm}} (V_{cm} - k_v \omega_{cp}) \quad (46)$$

with k_t , k_v and R_{cm} the motor constants, η_{cm} the motor mechanical efficiency and V_{cm} the motor voltage and the torque load τ_{cp} by:

$$\tau_{cp} = \frac{c_{p,air} T_{cp,in}}{\eta_{cp} \omega_{cp}} \left[\left(\frac{p_{cp,out}}{p_{cp,in}} \right)^{1-1/\gamma} - 1 \right] \dot{m}_{cp} \quad (47)$$

The compressor mass flow rate $\dot{m}_c$ is generally determined from the rotational speed and the pressure ratio p_r with the help of a compressor map, that can be obtained experimentally or from supplier data and fitted to non-linear functions [28,30,73]. Some authors [48,49,81,82] neglected the dynamics of the compressor and used only a static model, with the required mass flow rate being given by the cathode stoichiometry λ_{ca} and the stack current:

$$\dot{m}_{air}^{req} = \lambda_{ca} \frac{i A_{cell} N_{cells}}{4F} \frac{M_{air}}{x_{O_2}} \quad (48)$$

Others [53,63] considered that the mass flow rate followed the required mass flow rate with a first-order transfer function and a fixed time constant. The compressor isentropic efficiency η_{cp} can be obtained from the compressor map [29,48] or considered as a constant [39,49,63,81]. In addition to the compressor isentropic efficiency, mechanical transmission, DC motor and power converter efficiencies can also be considered [49,82]. It is important to note that the compressor isentropic efficiency will have a significant impact on the air temperature at the outlet of the compressor, which is given by [30]:

$$T_{cp,out} = T_{cp,in} + \frac{T_{cp,in}}{\eta_{cp}} \left[\left(\frac{p_{cp,out}}{p_{cp,in}} \right)^{1-1/\gamma} - 1 \right] \quad (49)$$

In Hoeflinger and Hofmann [51], the compressor pressure ratio and power consumption were fitted to non-linear functions of the mass flow rate and of the voltage applied to the air throttle valve placed downstream the cathode outlet. A review on different type of compressors and their application in fuel cell systems can be found in [117], which also included a short review on the mathematical modelling of air compressors.

In the case of open-cathode fuel cells, fans are generally used for the cathode air supply [118]. Therefore, fan models must be included in works dealing with open-cathode configurations, such as the one of Strahl et al. [36]. These authors adopted a polynomial fit of the air speed at the cathode inlet as a function of the fan duty cycle which was obtained experimentally. Further, they considered a first-order with a time constant of approximately 1 s to model the fan dynamics.

3.2. Air humidifier

The air humidifier ensures that the cathode air is sufficiently humid to avoid the dehydration of the membrane. In their review on humidification strategies for PEM fuel cells, Chang et al. [119] listed different types of air humidifiers: gas bubbling, direct water injection, enthalpy wheels, membrane humidifiers and exhaust gas recirculation. Due to their impact on the performance of the fuel cell system, air humidifier models were sometimes included in system models. Yang et al. [28] used a plane membrane humidifier model analogous to their fuel cell model, but with a single control volume along the channel. This model was validated against water vapour transfer rate measurements for two dry side inlet temperatures (50 °C and 60 °C). To our knowledge, it is the only model that takes into account the dynamics of the membrane water content, although it was validated against steady-state measurements only. Tang et al. [100] and Kang and Min [48] also adopted a membrane humidifier model analogous to their respective fuel cell model, but with more control volumes along the channel, as schematized in Fig. 4. Kim et al. [39] and Fu et al. [80] used a 0D model for the membrane humidifier accounting for heat and water transport across the membrane. A simpler approach adopted for instance by some authors [30,31,34,53,61,74] is to consider the air “ideally humidified” in the fuel cell inlet, thus avoiding any modelling of an air humidifier. In this case, the water mole fraction at the cathode inlet is readily calculated from the required relative humidity, the cathode inlet pressure and the saturation vapour pressure of water at the cathode inlet temperature $p_{sat}(T_{ca,in})$:

$$x_{H_2O,ca,in} = RH_{ca,in}^{req} p_{sat}(T_{ca,in}) / p_{ca,in} \quad (50)$$

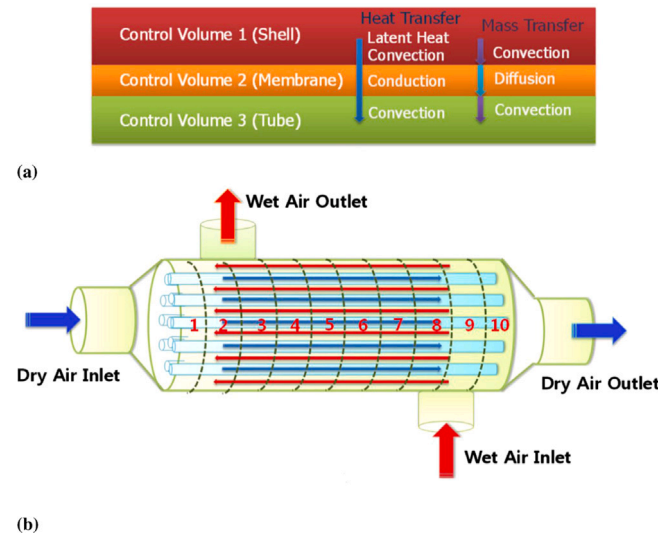


Fig. 4. Schematics of the 1D+1D shell-tube humidifier model used by Kang et al. [48]: (a) through-plane and (b) along-the-channel discretizations.

Source: Reproduced from Kang et al. [48] with permission from Elsevier.

Schröder et al. [49] also considered the air to be ideally humidified, but they accounted for the pressure drop of the membrane humidifier in their calculations. Fly [63] used an iterative method to calculate water and heat transfer in the air humidifier given a target relative humidity at the cathode inlet. De Campo [73] modelled an isothermal direct water injection by assuming that the dry gas partial pressure remain constant. Xu et al. [29] used an air humidifier map providing the dew point temperature at the outlet from the dry air mass flow rate and temperature. Membrane humidifier models can also be found in dedicated works, which proposed steady-state or dynamic 0D [120] or 1D models [121–123], as well as models based on the NTU-effectiveness method [124,125] or on experimentally obtained maps [126].

3.3. Anode recirculation

Anode recirculation is generally used to allow for a high hydrogen stoichiometry without increasing hydrogen waste. Anode recirculation is implemented with a hydrogen mechanical pump, an ejector or an electrochemical pump, plus a gas–water separator [127]. Karnik et al. [58,128] and Bao et al. [55] calculated the ejector primary and secondary inlets from supply and return manifold pressures and temperatures. Yang et al. [28] modelled an electrochemical hydrogen pump used to hydrogen recirculation, with a model analogous to the one used for the fuel cell or the membrane humidifier. A simpler approach was adopted by Ritzberger et al. [53] and Liso et al. [34] who assumed that the recirculation mass flow rate was known. In order to avoid the modelling of the anode recirculation loop, De Campo [73] used measurements of the anode inlet and outlet pressures and considered that the anode gas was saturated with water vapour in order to calculate the hydrogen partial pressure.

3.4. Cooling loop

In the case of liquid-cooled PEM fuel cells, the cooling loop ensures that the waste heat is evacuated from the fuel cell system. It generally consists of a coolant tank, a coolant pump and a radiator heat exchanger, often with a bypass [108]. Modelling the cooling loop allows to better account for the dynamics of the cooling fluid inlet temperature and, as a consequence, of the fuel cell stack temperature. Several authors [28,63] adopted NTU-based effectiveness models for the heat exchangers, while neglecting the heat capacity of the cooling fluid in

the circuit. Schultz and Horn [69] also used a NTU-based method but instead take into account the heat capacity of the coolant fluid. In a NTU-based method, the effectiveness of the radiator heat exchanger is obtained from the number of transfer units using a correlation, such as the one for a single-pass cross-flow heat exchanger with both fluids unmixed [129]:

$$\varepsilon = 1 - \exp \left\{ \frac{NTU^{0.22}}{C_r} \left[\exp(-C_r NTU^{0.78}) - 1 \right] \right\} \quad (51)$$

where:

$$NTU = UA/C_{min} \quad (52)$$

$$C_{min} = \min(\dot{m}_{cool}c_{p,cool}, \dot{m}_{air}c_{p,air}) \quad (53)$$

$$C_r = \frac{\min(\dot{m}_{cool}c_{p,cool}, \dot{m}_{air}c_{p,air})}{\max(\dot{m}_{cool}c_{p,cool}, \dot{m}_{air}c_{p,air})} \quad (54)$$

and UA is the overall heat transfer coefficient. Detailed calculations to obtain the overall heat transfer coefficient for louvered fin radiators typically used in automotive applications can be found in [28,63]. Once the effectiveness is obtained, the heat dissipated by the radiator is given by:

$$\dot{Q}_{rad} = \varepsilon C_{min}(T_{cool,in} - T_{air,in}) \quad (55)$$

In a different approach, Kang et al. [48] and Tang et al. [100] adopted 1D radiator models with respectively five and four control volumes in the flow direction. Wang et al. [79] adopted a similar model but with a single control volume in the flow direction, while accounting for the thermal inertia of the radiator walls. Liso et al. [34] used a linear parametrization of the heat exchanger overall heat transfer coefficient as a function of the air flow rate. The coolant tank can also be modelled, as done by Fly and Thring [130]. They used a single energy balance equation:

$$\dot{m}_{tank}c_{p,cool} \frac{dT_{tank}}{dt} = \dot{Q}_{cool} - \dot{Q}_{rad} \quad (56)$$

where $\dot{Q}_{cool}$ is the heat transferred from the stack to the coolant, $\dot{Q}_{rad}$ the heat dissipated by the radiator heat exchanger, $c_{p,cool}$ the coolant specific heat and m_{tank} the coolant mass in the tank. It should be noted that this formulation neglects the heat capacity of the walls of the coolant tank.

3.5. Condenser

Condensers can be present in some fuel cell systems where water recovery is important for water management, for instance in direct water injection humidification systems [131]. Condenser models with different degrees of complexity can be found in some works. Karnik et al. [58] modelled the condenser by a fixed liquid water removal efficiency. Striednig et al. [82] considered that the air at the exit of the condenser reached a desired temperature, with 100 % relative humidity and no liquid water. Fly and Thring [63,130] adopted an iterative model of the condenser which was able to predict both the heat transfer and the condensate rate. Their model was validated against experimental measurements. A 0D condenser model was also used by Schultze and Horn [60].

3.6. Inter-cooler

The inter-cooler is a heat exchanger placed after the compressor to reduce the air temperature before the air enters the air humidifier. Modelling the inter-cooler heat exchanger allows one to better describe the behaviour of the temperature at the inlet of the air humidifier, and therefore the performance of this component. However, only few authors considered the inter-cooler in their models. Kim et al. [59] assumed that the temperature exiting the inter-cooler downstream

the air compressor was constant and known. In Yang et al. [28] the temperature at the inlet of the dry side of the membrane humidifier was also considered to be constant and equal to 30 °C. Xing et al. [81] modelled the inter-cooler using a NTU-based method and a constant heat transfer coefficient. They showed that the inter-cooler corresponds to less than 1 % in the thermal load of the cooling circuit.

3.7. Piping and manifolds

Finally, the different system components are connected by pipes and inlet/exhaust manifolds. Usually the pipes and manifolds between two components are lumped into a volume that behaves like a perfectly stirred tank, as proposed by Pukrushpan et al. [30]. Schultze and Horn [69] are the only authors that considered the time delay due to the flow speed in the pipe between different system components.

4. Model validation

Validation of fuel cell models is fundamental, specially as several more or less strong hypotheses are made to reduce the model complexity and computational cost. Most often, measurements of stack voltage and temperature are used to validate the models as obtaining them is straightforward. But for water management, quantities such as the water membrane content, water transport across the membrane, liquid water saturation, high-frequency resistance, outlet relative humidities and their spatial distribution and dynamics are of great importance. If accessing those quantities *in situ* remain difficult, some authors were able to validate their models with interesting data. Robin et al. [44] compared their model results to 2D current density and temperature distributions across the cell surface for three humidity conditions. The authors stated that a good qualitative agreement was obtained, but noted significant differences due to effects that were not accounted for in the model, such as a better electronic contact near the edges of the cell due to mechanical compression and the bipolar plate welding. These effects were indeed shown to be significant by Liang et al. [132] and Peng et al. [133]. The validation with 1D along-the-channel current density measurements generally gives a better agreement, as evidenced by Yin et al. [134], since edge effects are smoothed by the integration along the direction perpendicular to the channels. Tang et al. [41] and Gong et al. [46] compared their model to 1D along-the-channel current density measurements, obtaining very good agreement. In a sequence of the work of Robin et al. [44], Nandjou et al. [10] and Tardy et al. [135,136] used the liquid water distribution over the cell surface area obtained with neutron imaging experiments to validate their numerical model. Simulation results of a 3D model were also compared to neutron imaging data by Hao et al. [137]. By providing detailed information on the spatial distribution of liquid water, neutron imaging experiments can be of great value for the development and validation of fuel cell models. Futter et al. [42] compared their model to electrochemical impedance spectra obtained at two different humidity conditions, but at a rather low current density of 0.2 A cm⁻². They suggested that the main deviations between their model and the experiments concerned a capacitive peak around 1 Hz and were due to the gas concentration gradients along the channel direction, which were related to gas channel bends and not well represented by their model. Other authors compared their models to measurements of the ohmic resistance [28,43,46,72] or the ohmic losses [11], which is an indicator of the membrane resistance and therefore of its degree of hydration. Also, Liso et al. [68] and Falcão et al. [38] used respectively the cathode outlet mass flow rate and the net water transfer coefficient across the membrane for validation, while Salva et al. [45] validated their model against measurements of the fuel cell water content. To our knowledge, Headley et al. [67] are the only authors who validated their model against transient measurements of the relative humidity at the cathode outlet.

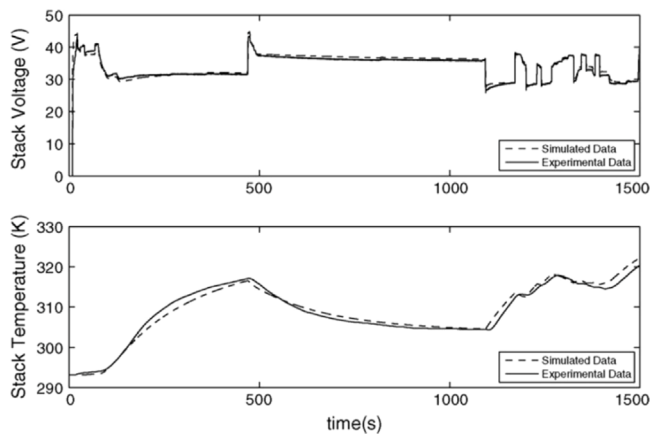


Fig. 5. Stack voltage and temperature results for a Ballard Nexa 1.2 kW fuel cell system under variable load: comparison between simulation and experimental data from Del Real et al. [31].

Source: Reproduced from Del Real et al. [31] with permission from Elsevier.

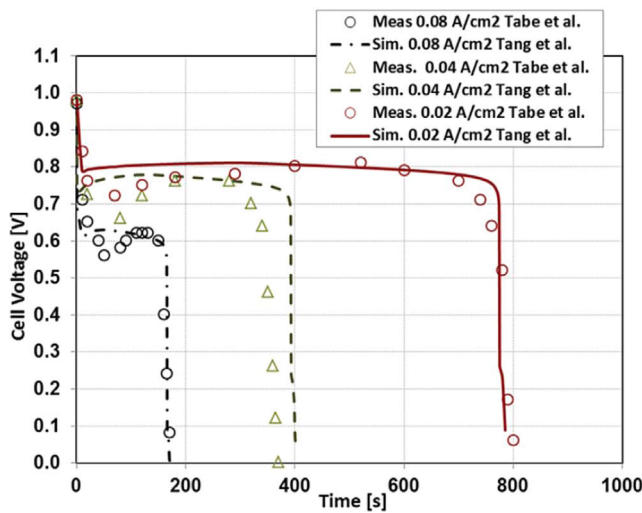


Fig. 6. Voltage break-in for the failed cold start of a fuel cell system: comparison between simulations from Tang et al. [41] and experimental data from Tabe et al. [140].

Source: Reproduced from Tang et al. [41] with permission from Elsevier.

More commonly, authors used transient stack/cell voltage and temperature data to validate their model. [28,29,31,36,41,43,46,53,57,60,72,73,77]. A good example is the work of del Real et al. [31] where the validation was based on transient voltage and temperature data for three representative situations (fuel cell startup, flooding and variable load) with a Ballard Nexa 1.2 kW fuel cell system. Fig. 5 shows the comparison for a variable load situation. Studies focused on cold start also used transient data for successful and failed fuel cell startup [28,35,41,46,57], such as the one in Fig. 6, extracted from Tang et al. [41]. Nevertheless, too many authors [33,34,48,50,55,63–65,80,81] compared their models to static voltage measurements or polarization curves only, even if they are supposed to simulate transient effects. Worse, some authors, even in recent studies, did not present any validation of their models at all [58,61,75,78]. This jeopardizes the reliability of fuel cell models, since several studies highlighted the need for a validation methodology as comprehensive as possible [11,138,139].

Some authors, compared different versions of their model to evaluate a given feature. Liso et al. [68] used this approach to show that considering non-equilibrium conditions for membrane water adsorption/desorption was needed to reproduce the experimental data. Futter

Table 2

Computational speed for different models as compared to real-time (RT).

1-5x faster than RT	5-50x faster than RT	>50x faster than RT
Goshstasbi et al. [66]	Salva et al. [45]	Khan and Iqbal [54]
Vetter and Schumacher [76]	Ritzberger et al. [53]	Headley et al. [67]
Gong et al. [46]	Goshstasbi et al. [43]	Schröder et al. [49]
	Xu et al. [29]	

et al. [42] compared a through-plane 1D version of their model with the original 1D+1D to show the impact of the along-the-channel gradients and to propose an explanation to the differences observed between their model and the EIS experimental data. The importance of the along-the-channel discretization was also shown in the work of Headley et al. [67] by comparing their 1D model to its 0D version with a single control volume. Strahl et al. [36] compared their model to a simpler version which does not take into account the impact of the saturation on the catalyst active area, showing that the simpler model cannot reproduce the behaviour observed experimentally. Beyond the validation of a model against experimental data, such a comparison between two (or more) versions of a model is a very good practice as it allows one to justify (or not) the need for a more complex model.

Concerning the validation of the models for other system auxiliaries, it was only present in some works as the focus is frequently on the fuel cell model. Kang et al. [48] validated their membrane humidifier and radiator models against steady-state experimental data. They used the dry side outlet dew point as a function of the air flow rate and temperature for the membrane humidifier and the heat rejection rate as a function of air and coolant mass flow rates for the radiator. Yang et al. [28] confronted their models for different system components with experimental data. For the membrane humidifier model, they used data for the water vapour transfer rates as a function of the air flow rate and temperature and for the water recovery ratio as a function of the pressure and temperature. For the radiator, they used data for the coolant outlet temperature for different coolant and air mass flow rates. For the electrochemical hydrogen pump, they used the polarization curves for different CO_2/H_2 ratios. Such an extensive validation could be used as a benchmark for future system models. However, in general, validation of models for fuel cell system components deserve more attention, particularly for the transient behaviour of components as the air humidifier or the radiator, which are fundamental for water and thermal management in most systems.

Another important aspect concerning validation is the use of benchmarks or standard conditions. Recently, Vetter and Schumacher [76] used operating conditions established by the Joint Research Centre of the European Commission for automotive applications in an attempt to establish a baseline for model benchmark and validation. But the fuel cell modelling community still lacks a benchmark for model validation, as highlighted by the recent review of Zhao et al. [16], despite a long effort for the development of fuel cell models. The fact that few models are available as open-source also hinders their validation by the community, as explained by Weber et al. [14], as well as cross-validations between different models and the definition of a benchmark or reference model. Once again, Vetter and Schumacher [76] tried to tackle the issue by proposing their model as an open-source, portable MATLAB file.

5. Computational time

Unfortunately, only few studies mentioned the computational time needed to run the proposed models. Yang et al. [28] only mention that its model was implemented in Matlab 2018a and run with a "powerful workstation". But after simplifications done by Gong et al. [46] their model, implemented in C, could be run 2 to 3 times faster than real-time on a computer with a 2.4 GHz processor. Goshtasbi et al. [43] implemented their model in C but used it as an S-Function in Simulink. They

mentioned that their model runs about 50 times faster than real-time also on a computer with a 2.4 GHz processor for co-flow configurations and 2 or 3 times slower than that for counter-flow configurations, due to the iterative solution. The steady-state model of Salva et al. [45] took less than 2 s to solve for around 20 steady-state operating points, which would make it 10 times faster than real time in a quasi-steady approach with a time step of 1 s. Schröder et al. [49] instead claimed that their steady-state model can simulate several hundred points per second, while the model proposed by Vetter and Schumacher [76] would simulate only five points per second with the same computer. Xu et al. [29] mentioned that their model required 18 s to simulate 500 s of operation, making it around 30 times faster than real-time, whereas an equivalent 1D model would be 60 times slower. Ritzberger et al. [53] claimed their model runs around 10 times faster than real time in a single-core personal laptop with a 2.6 GHz processor. The model of Headley et al. [67] took around 100 s to simulate 160 min with the along the channel discretization with six control volumes and ten times less in its 0D version with a single one. The fastest computational time reported is the one of Khan and Iqbal [54], who stated that their model runs 3000 times faster than real-time. The performances of those models are summarized in Table 2. It should be highlighted that those models have different levels of complexity and accuracy and have been designed to address different problems. That explains the wide range of computational times reported.

6. Summary and perspectives

Throughout this review, we attempted to identify the modelling approaches and hypotheses commonly adopted in physics-based proton-exchange membrane fuel cell models that could be useful for system-level water and thermal management studies. We put in evidence the high diversity of approaches adopted in the literature concerning the cell voltage and the different overpotentials, the transport of gases, liquid water and heat inside the cell, the spatial heterogeneity at the cell or the stack level and the dynamics of different phenomena. This diversity leads to models with different levels of computational time (ranging from 1x to more than 50x faster than real-time), complexity and accuracy. But comparing the different models and modelling approaches in an objective way remain a very challenging task. Unfortunately, numerical implementations of most models are not openly available. Further, only very few works indicate clearly the computational time required to perform simulations with their models. Worse no standard validation benchmark exist. As system-level studies require models that are computationally efficient and validated against relevant experimental data, we believe that the development of standard validation procedure is an urgent matter. While most models described in this work concern only the fuel cell stack, some authors also present models for the system components, such as the compressor, the air humidifier or the cooling circuit radiator. Once again, in only few cases those models are validated.

Given the diversity of models already available in the literature, rather than developing new models from scratch, it would be more appropriate to select an existing model, perhaps adding some incremental changes. These incremental changes could then be verified by comparing the results with the ones of the original model. The selection of the existing model depends on the modelling goals and data availability. For instance, if predictive capabilities are required and detailed information about the cell materials are available, the model of Yang et al. [28] could be selected. The slip ratio parametrization proposed by Tang et al. [41] could be added to that model if liquid water transport in the gas channels is expected to be important. The consideration of platinum oxide growth dynamics proposed by Goshtasbi et al. [43] could be added if low-current regions are of interest. The model of Gong et al. [46] is an alternative if computational cost is an issue. If low information about the cell materials is available and the goal is to fit a computationally efficient model to operational data, the models

developed by Schröder et al. [49] or Ritzberger et al. [27] may be the best. The model of Goshtasbi et al. [43] has a quite complex description of the catalyst layer and a high number of fitted parameters, while the interest of such complexity remains to be demonstrated. The network topology of the model proposed by Tang et al. [41] does not seem simple to implement at first sight. For steady-state investigations where detailed cell material data is available, Vetter and Schumacher [76] open model is an immediate choice. The model of Xu et al. [29] which uses analytical solutions of water and gas transport in the GDL can be a very interesting compromise between the capture of two-phase water dynamic effects and computational cost. However, its use requires further validation on more diverse operating conditions. The model of Jiang et al. [11] can be used instead if dynamic effects are not of importance. Finally, the model of Xing et al. [81] is of interest when few data is available and the user is interested in thermal management at normal operating conditions (fully humidified membrane). To sum up, if several models with different levels of computational time, complexity and accuracy are available in the literature, significant efforts are still needed to understand the impact of different modelling approaches and hypotheses. This can only be reached by extensive validation of the developed models against experimental data and against each other. That will lead to more reliable and efficient proton-exchange membrane fuel cell models, and as a consequence, more reliable and efficient system-level studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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